

The oneirogen hypothesis: modeling the hallucinatory effects of classical psychedelics in terms of replay-dependent plasticity mechanisms

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eLife Assessment

This paper provides a **useful** new theory of the hallucinatory effects of psychedelics. The authors present **convincing** evidence that a computational model trained with the Wake-Sleep algorithm can reproduce some features of hallucinations by varying the strength of top-down connections in the model, but discussion of the model's relationships to the psychedelics and sleep literatures is **incomplete**. The work will be of interest to researchers studying hallucinations or offline activity and learning more broadly.

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Abstract

Classical psychedelics induce complex visual hallucinations in humans, generating percepts that are co-herent at a low level, but which have surreal, dream-like qualities at a high level. While there are many hypotheses as to how classical psychedelics could induce these effects, there are no concrete mechanistic models that capture the variety of observed effects in humans, while remaining consistent with the known pharmacological effects of classical psychedelics on neural circuits. In this work, we propose the “oneirogen hypothesis”, which posits that the perceptual effects of classical psychedelics are a result of their pharmacological actions inducing neural activity states that truly are more similar to dream-like states. We simulate classical psychedelics’ effects via manipulating neural network models trained on perceptual tasks with the Wake-Sleep algorithm. This established machine learning algorithm leverages two activity phases, a perceptual phase (wake) where sensory inputs are encoded, and a generative phase (dream) where the network internally generates activity consistent with stimulus-evoked responses. We simulate the action of psychedelics by partially shifting the model to the ‘Sleep’ state, which entails a greater influence of top-down connections, in line with the impact of psychedelics on apical dendrites. The effects resulting from this manipulation capture a number of experimentally observed phenomena including the emergence of hallucinations, increases in stimulus-conditioned variability, and large

increases in synaptic plasticity. We further provide a number of testable predictions which could be used to validate or invalidate our oneirogen hypothesis.

1 Introduction

Classical psychedelics—including psilocybin, mescaline, DMT, and LSD—are a family of hallucinogenic compounds with a common mechanism of action: they are agonists for the 5-HT_{2a} serotonin receptor commonly expressed on the apical dendrites of cortical pyramidal neurons [1]. These drugs induce numerous effects in human subjects, including: complex visual, auditory, and tactile hallucinations; intense spiritual experiences; long-lasting alterations in mood; changes in personality; and increases in synaptic plasticity [2, 3, 4]. They have further been used for millennia as medicine and in religious rituals [5]; more recently, they have been explored clinically as potential treatments for depression and anxiety [6], as well as PTSD [7].

The 5-HT_{2a} receptor plays a critical role in psychedelic-induced hallucinations. Indeed, perceptual effects are largely eliminated by blocking these receptors in the cortex [8, 9]. However, very little is understood about *why* highly structured hallucinations and changes in synaptic plasticity emerge from activating cortical 5-HT_{2a} receptors: to explain this, it is necessary to develop mechanistic theories that are capable of linking changes in neuron-level properties (receptor agonism) to changes in perception and behavior. Psychedelic drug users and therapists have long noted the ‘dream-like’ qualities of psychedelic drug hallucinations, which are realistic but untethered from the external world; this observation leads naturally to speculation that these drugs are ‘oneirogens,’ or dream-manifesting compounds [10]. However, beyond perceptual phenomenology (and some evidence pointing to the effects of psychedelics on sleep cycles [11, 12, 13]), we lack a mechanistic proposal that could explain the similarity between dreams and psychedelic drug experiences. Here, we articulate the ‘oneirogen hypothesis’, which describes one such potential mechanistic explanation. We propose that classical psychedelics induce a dream-like state by shifting the balance between bottom-up pathways transmitting sensory information and top-down pathways ordinarily used to create replay sequences in the brain. Replay sequences have been shown to be important for learning during sleep [14, 15, 16, 17, 18]: we propose that mechanisms supporting replay-dependent learning during sleep are key to explaining the increases in plasticity caused by psychedelic drug administration. In total, our model of the functional effect of psychedelics on pyramidal neurons could provide a explanation for the perceptual psychedelic experience in terms of learning mechanisms for consolidation during sleep [19], and cortical ‘replay’ phenomena [20, 21, 22, 23, 24, 25, 26, 27, 28].

To explore the oneirogen hypothesis concretely, we use the aptly named Wake-Sleep algorithm [29], which has historically been used to train artificial neural networks (ANNs) that possess both a bottomup “recognition” pathway and a top-down “generative” pathway to learn a representation of incoming sensory data. It enables unsupervised learning in ANNs by alternating between periods of “waking perception” (wherein bottom-up recognition pathways drive activity) and “dreaming sequences” (wherein top-down generative pathways drive activity). With these alternate periods of distinct activity, connectivity parameters in each pathway are adjusted to match the activity of the opposite pathway. This way, the top-down pathway learns to generate activity consistent with that induced by sensory inputs, and the bottom-up pathway learns better representations thanks to generated activity.

In this work, we show that within a neural network trained via Wake-Sleep, it is possible to model the action of classical psychedelics (i.e. 5-HT_{2a} receptor agonism) by shifting the balance during the wake state from the bottom-up pathways to the top-down pathways, thereby making the ‘wake’ network states more ‘dream-like’. Specifically, we model the effects of classical psychedelics by manipulating the relative influence of top-down and bottom-up connections in neural networks

trained with the Wake-Sleep algorithm on images. Doing so, we capture a number of effects observed in experiments on individuals under the influence of psychedelics, including: the emergence of closed-eye hallucinations, increases in stimulus-conditioned variability, and large increases in synaptic plasticity. This data suggests that the oneirogen hypothesis may indeed help to explain why 5-HT_{2A} agonists have the functional effects that they do. We subsequently identify several testable predictions that could be used to further validate the oneirogen hypothesis.

2 Results

Mapping the Wake-Sleep algorithm onto cortical architecture

The Wake-Sleep algorithm allows ANNs to optimize a global, unsupervised objective function for sensory representation learning—the Evidence Lower Bound (ELBO)—through local synaptic modifications to a bottom-up recognition pathway and a top-down generative pathway. As a precursor to the variational autoencoder [30, 31], the Wake-Sleep algorithm provides a mechanism for learning a probabilistic latent representation $\mathbf{r}$ responding to incoming sensory stimuli $\mathbf{s}$, which obeys representational characteristics that are ideal for a neural system (e.g. sparsity and metabolic efficiency [32], compression and coding efficiency [33, 34], or disentanglement [35, 36]). To do this, Wake-Sleep optimizes the ELBO through an approximation of the Expectation Maximization (EM) algorithm [37] to train the two pathways (Figure 1a). For readers who are unfamiliar with the Wake-Sleep algorithm, a tutorial can be found here [38].

Notably, the Wake-Sleep algorithm requires two phases of activity (i.e. “Wake” and “Sleep”), where the network phase is controlled by a global state variable $\alpha \in [0, 1]$ that regulates the balance between the bottom-up and top-down pathways. In the Wake phase ($\alpha = 0$), the network processes real sensory stimuli drawn from the environment, and network activity is sampled based on the bottom-up inputs (corresponding to the approximate inference distribution). In the Sleep phase ($\alpha = 1$), the network internally samples neural activity from its generative model, which then produces generated activity in the stimulus layer $\mathbf{s}$. We use this structure of the Wake-Sleep algorithm as a concrete model to express oneirogen hypothesis. Specifically, we use changes to the value of α as a means of modeling a 5-HT_{2A} agonist-induced shift to a more dream-like state, as we detail below.

Within the Wake-Sleep algorithm, neurons alternate between ‘Wake’ and ‘Sleep’ modes, where activity during each mode is dominated by the bottom-up and top-down pathways, respectively. We can determine the neural activity for a given intermediate layer l with the following equation:

$$\mathbf{r}^{(l)} = f(h(\mathbf{r}), \mu(\mathbf{r}), \alpha) + f(\sigma_b, \sigma_p, \alpha)\boldsymbol{\eta}, \quad (1)$$

where $h(\mathbf{r})$ defines bottom-up input, $\mu(\mathbf{r})$ defines top-down input, $f(h, \mu, \alpha)$ is any interpolation function such that $f(h, \mu, 0) = h$ and $f(h, \mu, 1) = \mu$, σ_b and σ_p define the bottom-up and top-down activity standard deviations, and $\boldsymbol{\eta} \sim \mathcal{N}(0, 1)$ adds random noise to the neural activity (see Methods for more detail). Here, for notational conciseness we treat $\mathbf{r}$ as a concatenated vector of all $\mathbf{r}^{(l)}$ vectors from each layer. This equation means that α controls whether bottom-up inputs or top-down inputs control the dynamics of individual neural units.

Thus, as α moves from a value of 0 to a value of 1 the activity of the neurons shifts from being driven by the bottom-up recognition pathway to being driven by the top-down generative pathway. How could this occur in the brain? In the cortex, excitatory pyramidal neurons receive inputs from distinct sources: inputs that are from ‘higher order’ cortical areas target the apical dendrites, whereas inputs that are from ‘lower order’ cortical or sensory subcortical areas target the basal dendrites [39]. Thus, we can capture the core idea behind the oneirogen hypothesis

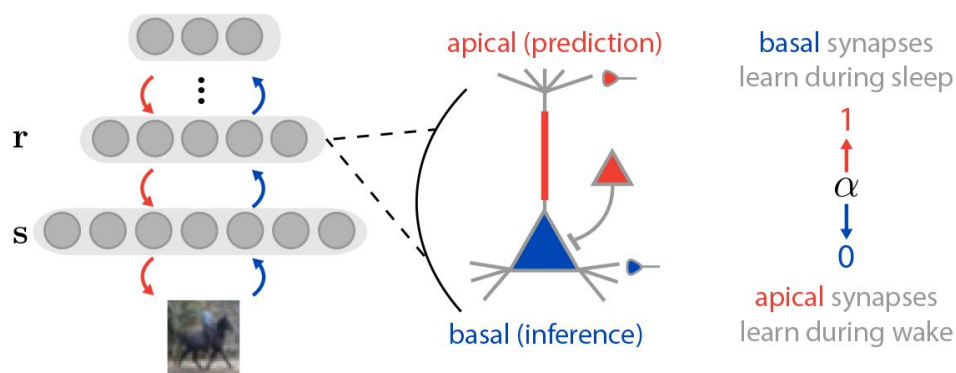


Figure 1.

Mapping the Wake-Sleep algorithm onto cortical architecture.

Left: Network architecture. We model early sensory processing in the cortex with a multilayer network, r , receiving stimuli s . Center: individual pyramidal neurons receive top-down inputs (red) at the apical dendritic compartment, and bottom-up inputs at the basal dendritic compartment (blue). 5-HT_{2a} receptors are expressed on the apical dendritic shaft (red bar), and on parvalbumin interneurons (red triangle); both sites may play a role in gating basal input. Right: Over the course of Wake-Sleep training, basal inputs dominate activity during the Wake phase ($\alpha = 0$) and are used to train apical synapses, whereas apical inputs dominate activity during the Sleep phase ($\alpha = 1$) and are used to train basal synapses.

using the Wake-Sleep algorithm, by postulating that the bottom-up basal synapses are predominantly driving neural activity during the Wake phase (when α is low), while top-down apical synapses are predominantly driving neural activity during the Sleep phase (when α is high; **Figure 1**) [40]; this is in agreement with several recent theoretical studies that have proposed that apical dendrites could serve as a site for integrating top-down learning signals [41, 42, 43, 44, 45, 46], particularly those which propose that the top-down signal corresponds to a predictive or generative model of neural activity [47, 48]. The idea that apical dendritic influence increases during sleep (or replay) in cortex has been only partially supported by experiments [49], and is one critical testable prediction of our model. Notably, 5-HT_{2a} receptors are expressed in the apical dendrites of pyramidal neurons [1] and basal dendrite-targeting parvalbumin inhibitory interneurons [50], and have an excitatory effect that positively modulates glutamatergic transmission [51, 52]. These data suggest that 5-HT_{2a} agonists could have a push-pull effect on cortical pyramidal neurons, increasing the relative influence of apical dendrites and decreasing the relative influence of basal dendrites. Hence, we can model these effects by changing the α value in a Wake-Sleep trained network, and then ask whether the networks exhibit other phenomena that match the known impact of classical psychedelics on neural activity. We note that with this mapping of the Wake-Sleep algorithm to models of basal and apical processing, synaptic modifications at both apical and basal synapses correspond to minimizing a local prediction-error between top-down and bottom-up inputs (see Methods).

Modeling Hallucinations

To see whether a transition from waking to a more dream-like state would induce hallucinatory effects in our model, we trained multilayer neural networks with branched dendritic arbors (see Methods) on the MNIST digits dataset [53] using the Wake-Sleep algorithm and subsequently simulated hallucinatory activity by varying α (see Methods; Eq. (8)). We could visualize the effects of our simulated psychedelic with snapshots of the stimulus layer $\mathbf{s}$ at a fixed point in time for various values of α (**Figure 2**; see Supplemental Materials for videos). As α increased, we observed that network activity gradually deformed away from the ground-truth stimulus in a highly structured way, adding strokes to the original digit that were not originally present. At the highest values of α tested, we found that network states were wholly divorced from the ground-truth stimulus, but retained many characteristics of the MNIST digits on which the network was trained (e.g. smooth strokes and the rough form of digits). These results emphasize that hallucinations induced by a shift to a more dream-like state in these models are heavily influenced by the training dataset, which for an animal would correspond to the statistics of the sensory environment in which it learns its sensory representation. To emphasize this point, we further trained our networks on the CIFAR10 natural images dataset [54] (**Figure 2c**), to provide an example of a more naturalistic training dataset. In this case, our model was not powerful enough to reproduce realistic natural images—instead, we found that our modeled hallucinatory activity corresponded to ‘ripple’ effects, which are similar to the ‘breathing’ and ‘rippling’ phenomena reported by psychedelic drug users at low doses [2].

These simulations were produced with a complex, multicompartmental neuron model; however, we found similar results with two alternative network architectures, one with within-layer recurrence (**Supplemental Figure S1a**) and one which used a simpler single compartment neuron model (**Supplemental Figure S1b**). We found that our single compartment model produced qualitatively less realistic generated images than the multicompartment and recurrent models, justifying our use of the more complex models (**Supplemental Figure S2**). To demonstrate the importance of a learned top-down pathway to produce complex, structured hallucinations in the earliest layers of our network, we generated model hallucinations from two control networks: an untrained model and a trained network where psychedelic activity was alternatively modeled by a simple increase in the variance of individual neurons (we will refer to this latter control as the noise-based hallucination protocol). We found that hallucinations under these control conditions resembled additive white noise, rather than structured digit-like shapes (**Supplemental Figure S1c-d**).

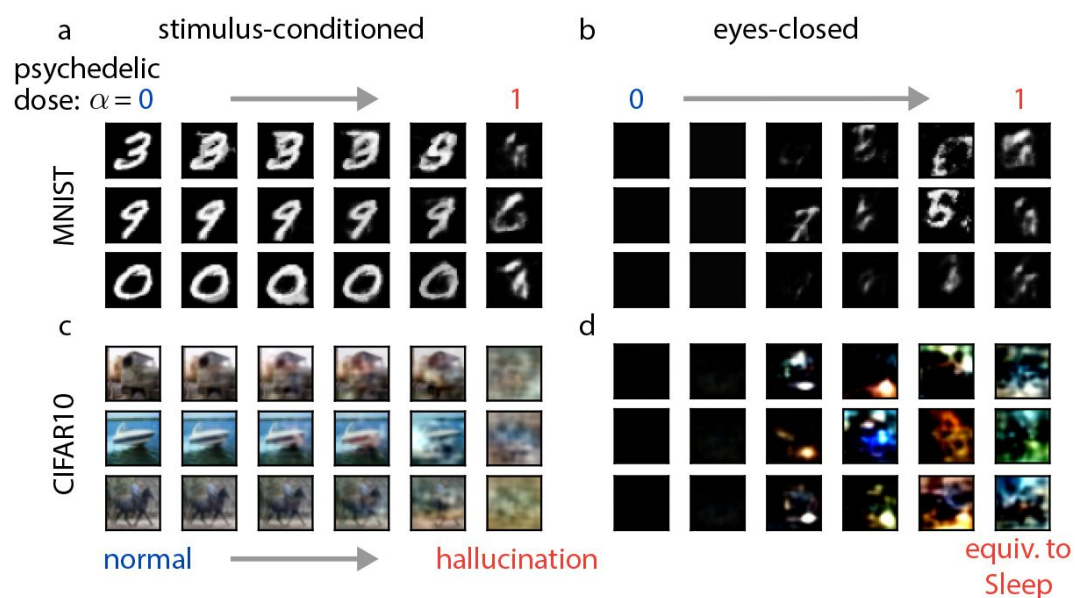


Figure 2.

Visualizing the effects of psychedelics in the model.

We model the effects of classical psychedelics by progressively increasing α from 0 to 1 in our model, where $\alpha = 1$ is equivalent to the Sleep phase. We visualize the effects of psychedelics on the network representation by inspecting the stimulus layer **s**. **a**) Example stimulus-layer activity (rows) in response to an MNIST digit presentation as psychedelic dose increases (columns, left to right). **b**) Same as (a) but for 'eyes-closed' conditions where an entirely black image is presented. **c-d**) Same as (a-b), but for the CIFAR10 dataset.

Psychedelic drug users also report observing the emergence of hallucinations while their eyes are closed [2]. Interestingly, we found that our model recapitulated these phenomena: as α increased, networks trained on MNIST gradually began revealing increasingly complex and digit-like patterns (Figure 2b), whereas CIFAR10-trained networks again predominantly produced ‘ripple’ hallucinations (Figure 2d).

Effects of psychedelics on single neurons

Having recapitulated hallucinatory phenomena in stimulus space, we next explored how our proposed mechanism affected neural activity in our network model, in order to establish markers that could be used to experimentally validate or invalidate the oneirogen hypothesis. To start, we investigated the effects of learning and psychedelic drug administration on the activity of single neurons in the model. As noted previously, the learning algorithm used here trains synapses so that top-down inputs to apical dendritic compartments match bottom-up inputs to basal dendritic compartments. As a consequence, we observed that after training, inputs to apical and basal dendritic compartments were much more correlated on the same neuron than they were for random neurons (Figure 3a), which was not observed in untrained models (Supplemental Figure S3a). This form of strongly correlated tuning has been observed in both cortex and the hippocampus [55, 56].

There are many indicators that psychedelic drug administration in humans and animals can induce marked, long-lasting changes in behavior, as well as large increases in synaptic plasticity [3, 57, 58, 9, 4]. In Wake-Sleep learning, apical synapses learn during the Wake phase, whereas basal synapses learn during the Sleep phase—thus, plasticity at apical synapses is gated by $(1 - \alpha)$, whereas plasticity at basal synapses is gated by α (see Methods). However, learning is still theoretically possible without this explicit gating, though it may be noisier and less efficient; furthermore, it is conceivable that classical psychedelics could increase the relative influence of apical inputs on the activity of a neuron without affecting this gating mechanism. As a consequence, we modeled the dose-dependent effects of psychedelics on plasticity both with and without gating (Figure 3b). Consistent with recent experimental results [3], for intermediate doses we found large increases in plasticity at both apical and basal synapses under *both* conditions, where plasticity was measured as a mean change in normalized synaptic strength across weight parameters in our network (see Methods). In our model, we found that the total evoked plasticity peaked at roughly $\alpha = 0.5$; we further found that if gating was affected by psychedelics, apical plasticity would eventually be quenched at very high drug doses. We also found that plasticity induced by psychedelic drug administration gradually became unaligned from the weight updates that would have occurred in the absence of the drug (Figure 3c), indicating that these results were not simply due to modulation of the effective learning rate of the underlying plasticity. Rather, as has been suggested by other theoretical studies [59], plasticity in the model likely increased because aberrant hallucinatory activity pulled the learning mechanism out of a local optimum in which plasticity was minimal, producing much more plasticity across the network. Importantly, we observed these increases in plasticity in all network architectures and training datasets we explored, including for our noise-based hallucination protocol (Supplemental S4), demonstrating that changes in apical dendritic influence within a Wake-Sleep learning framework are sufficient, but *not necessary* to induce increases in synaptic plasticity: for trained networks, it would seem that even simple increases in neural variability can have similar effects.

Effects of psychedelics on neural variability

Having observed that increasing our modeled drug dosage caused heightened fluctuations and deviations from the ground-truth stimulus in the sensory layer of our network (Figure 2), we next investigated whether variability was affected at the level of individual neurons in higher layers of the model. Indeed, we found that for a fixed stimulus, neural variability increased markedly as the simulated psychedelic drug dose increased (Figure 4a). This result is consistent

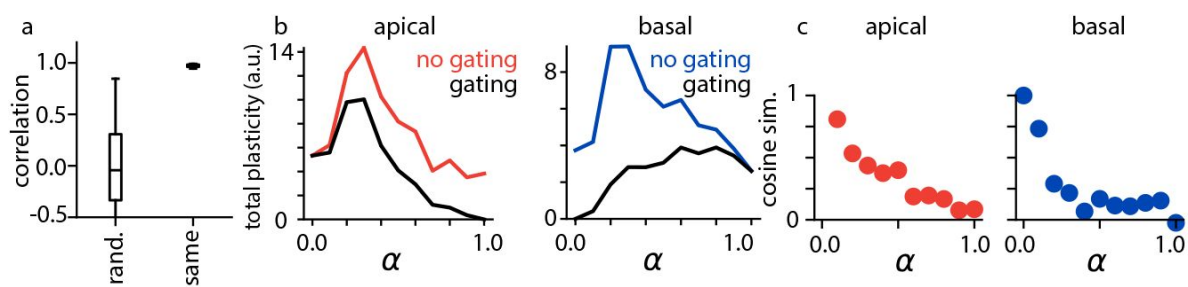


Figure 3.

Effects of psychedelics on single model neurons.

a) Correlations between the apical and basal dendritic compartments of either the same network neuron or between randomly selected neurons. Total plasticity for apical (left) and basal (right) synapses as α increases in the model when plasticity is either gated or not gated by α . Error bars indicate ± 1 s.e.m. **c)** Cosine similarity between plasticity induced under psychedelic conditions compared to baseline for apical (left) and basal (right) synapses.

with the data supporting the Entropic Brain Theory [60, 61, 62, 63], in which neural activity in resting state fMRI recordings becomes increasingly ‘entropic’ (i.e. variable) under the influence of psychedelics; however, it is important to note that our noise-based hallucination protocol also produced these effects (**Supplemental Figure S5a**). Though most experimental data supporting the Entropic Brain Theory is taken from recordings with relatively poor spatial resolution, averaging activity over large cortical areas, our model predicts that this increase in variability should be reflected at the level of individual neurons; this increase in variability after psychedelic administration has been recently observed in auditory cortical neurons for active mice [64], but whether this phenomenon is general across tasks and cortical areas remains to be seen. We further found that this increase in variability corresponded to a decrease in ability to identify the stimulus being presented to the network: we trained a classifier to identify which MNIST digit was presented to our networks on Wake neural activity (see Methods), and found that the accuracy of our classifier decreased (**Figure 4b**) while the output variability of the classifier increased (**Figure 4c**) in response to drug administration.

Within our model, this increase in variability is quite sensible: in the ordinary Wake state, neural activity is constrained to correspond to the singular sensory stimulus being presented, whereas during Sleep states, neural activity is completely unconstrained by any particular sensory stimulus, reflecting instead the full distribution of *possible* sensory stimuli. As increasing α in our model interpolates between Wake and Sleep states, we can expect intermediate values of α to produce network states which are less constrained by the particular sensory stimulus being presented, reflected in increased neural variability.

Network-level effects of psychedelics

We next investigated the effects of psychedelics on network-level and inter-areal dynamics within our model. We first identified an important negative result: the pairwise correlation structure between neurons was largely preserved across psychedelic doses (**Figure 5a-b**), as was the effective dimensionality of population activity (**Figure 5c**). This was sensible, because a network that has been well-trained with the Wake-Sleep algorithm will have the same marginal distribution of network states in the Wake mode as in the Sleep mode—thus, pairwise correlations between neurons should also not differ (as measures of the second order moments of the marginal distribution). We found empirically that even for intermediate values of α in which activity is a mixture of Wake and Sleep modes, these correlations are largely unchanged; in contrast, we observed large changes in correlation structure for untrained networks, and increases in effective dimensionality for both untrained networks and for our simple noise-based hallucination protocol, suggesting that these results are more specific to our trained models in which hallucinations are caused by an increase in apical dendritic influence (**Supplemental Figure S6a-b**). Interestingly, these results are consistent with a recent study that has shown only minimal functional connectivity and effective dimensionality changes in task-engaged humans being presented audiovisual stimuli under the influence of psilocybin [63].

However, though the pairwise correlations between single neurons are largely preserved, the causal influence between lower and higher layers of our model network changes considerably both during hallucination and Sleep modes. Because psychedelic drug administration increases influence of apical dendritic inputs on neural activity in our model, we found that silencing apical dendritic activity reduced across-stimulus neural variability more as the psychedelic drug dose increases (**Figure 5d**). Further, we found that as α increased, inactivating the deepest network layer induced a large reduction in variability in the stimulus layer relative to baseline (**Figure 5e**), revealing that within our model, increases in top-down influence are responsible for much of the observed stimulus-conditioned variability at larger drug doses. These inactivations had no impact on neural variability in our noise-based hallucination protocol, but were observed for all network architectures and datasets that we tested in which hallucinations were caused by an

Figure 4.

Effects of psychedelics on neural variability.

a) Stimulus-conditioned variability for neurons in the network as α increases, as compared to variability in neural activity across stimuli (rightmost bar). Error bars indicate ± 1 s.e.m. **b)** Proportion correct for a classifier trained to detect the label of presented MNIST digits as α increases. **c)** Variability in the logit outputs of the trained classifier as α increases.

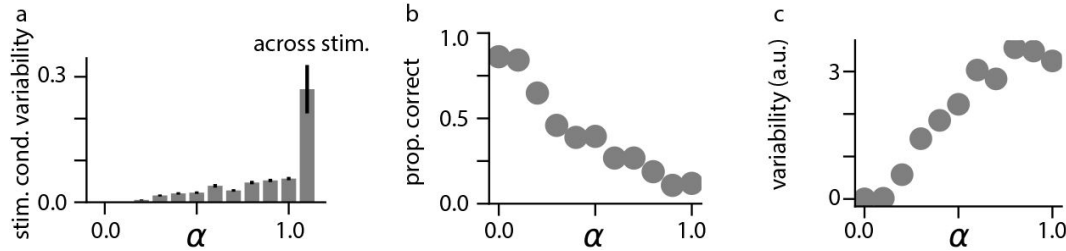
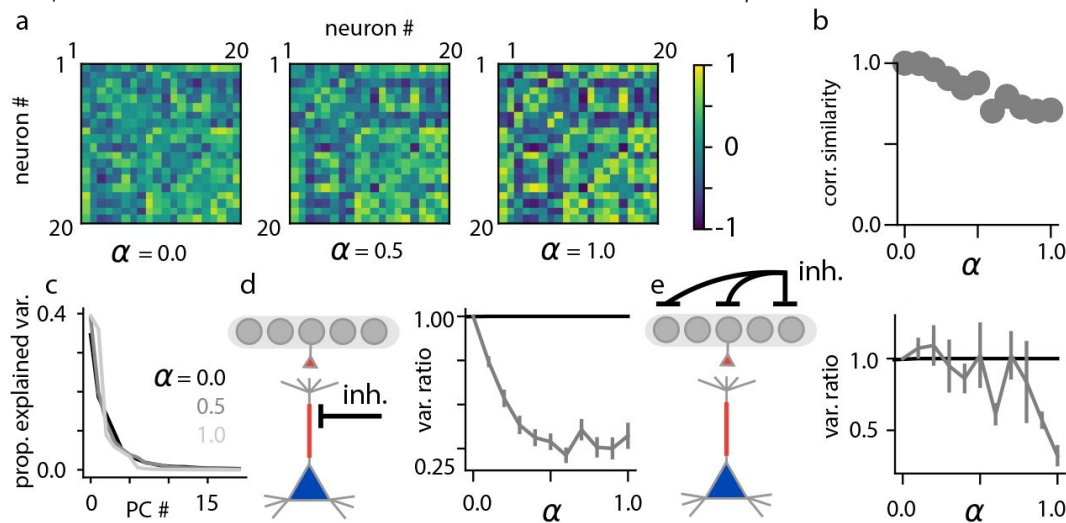


Figure 5.

Network-level effects of psychedelics.

a) Pairwise correlation matrices computed for neurons in layer 2 across stimuli for $\alpha = 0$ (left), $\alpha = 0.5$ (center), and $\alpha = 1.0$ (right). **b)** Correlation similarity metric between the pairwise correlation matrices of the network in the absence of hallucination ($\alpha = 0$) as compared to hallucinating network states ($\alpha > 0$). **c)** Proportion of explained variability as a function of principal component (PC) number for $\alpha \in \{0, 0.5, 1\}$. **d)** Ratio of across-stimulus variance in individual stimulus layer neurons when the apical dendrites have been inactivated, versus baseline conditions across different α values. **e)** Ratio of across-stimulus variance in individual neurons in the stimulus layer when neurons at the deepest network layer have been inactivated, versus baseline conditions across different α values. Error bars indicate ± 1 s.e.m.



increase in apical dendritic influence (**Supplemental Figure S6** [↗](#)), suggesting that these results are quite specific to our model. Further, these inactivations have not yet been performed in animals, and consequently constitute a critical testable prediction of our model.

3 Discussion

Experimental results captured by our model

In this study, we have examined a hypothetical mechanism explaining how the 5-HT_{2a} receptor agonism of classical psychedelics could induce the highly structured hallucinations reported by people who have consumed these drugs. Specifically, we have explored the ‘oneirogen hypothesis’, which postulates that 5-HT_{2a} agonists have the effects that they do because they shift the neocortex to a more dream-like state, wherein activity is more strongly driven by top-down inputs to apical dendrites than normally occurs during waking. To provide a concrete model to explore the ‘oneirogen hypothesis’ we used the classic Wake-Sleep algorithm, which learns by toggling between a Wake phase, where activity is driven by bottom-up sensory inputs, and a Sleep phase, where activity is driven by top-down generative signals. We modeled the ‘oneirogen hypothesis’ by simulating psychedelic administration as an increase in a neuronal state variable that switches neural activity between these two phases (α), such that the simulated psychedelic caused the network to enter a state somewhere between the Wake and Sleep phases, making activity during the Wake phase less tied to actual sensory inputs by increasing the relative influence of the top-down, apical compartment in the models (depending on the “dosage”). This formulation is consistent with anatomical wiring data [\[39\]](#), as well as several recent theoretical studies which propose a specialized learning role for top-down projections to the apical dendrites of pyramidal neurons [\[41\]](#), [\[42\]](#), [\[43\]](#), [\[44\]](#), [\[45\]](#), [\[46\]](#). It is also consistent with the known cellular mechanism of action of classical psychedelics [\[1\]](#), [\[51\]](#), [\[52\]](#), [\[8\]](#). Using this model, we were able to produce both stimulus-conditioned and “closed-eye” hallucinations that are consistent with the low-level effects reported by psychedelic drug users [\[2\]](#), and we were also able to recapitulate the large increases in plasticity observed at both apical and basal synapses at moderate psychedelic doses [\[3\]](#).

Our model uses a particular functional form of synaptic plasticity at both apical and basal synapses, reminiscent of the classical delta rule [\[65\]](#), which seeks to minimize a prediction error between inputs in apical and basal synapses. There are many theoretical models of learning that propose similar forms of plasticity [\[42\]](#), [\[43\]](#), [\[47\]](#), so while this plasticity is a necessary prediction of our model, it is not *sufficient* to validate it. Experimentally, plasticity dynamics which could, theoretically, minimize such a prediction error have been observed in cortex [\[66\]](#), [\[67\]](#), and it has also been proposed that behavioral timescale plasticity in the hippocampus could subserve a similar function [\[68\]](#). We found that plasticity rules of this kind induce strong correlations between inputs to the apical and basal dendritic compartments of pyramidal neurons, which has been observed in the hippocampus and cortex [\[55\]](#), [\[56\]](#).

Interestingly, we also found that increasing the influence of apical dendrites in the model increased stimulus-conditioned variability in our individual neurons. In cortex, this effect has recently been shown at the level of single auditory neurons [\[64\]](#); further, there have been numerous studies reporting similar increases in variability (or, analogously, entropy) in resting-state human brain recordings, previously modeled using Entropic Brain Theory [\[62\]](#). This theory proposes that many of the effects of classical psychedelics on perception and learning can be explained in terms of increases in variability induced by drug administration (e.g. the increase in variability could introduce novel patterns of thinking, or perturb learning to allow it to break out of ‘local minima’). Our results are broadly consistent with this perspective, to which we have added explanatory layers that are both normative and mechanistic [\[69\]](#), [\[70\]](#): namely, we speculate that this variability under ordinary conditions results from an ethologically important

mechanism underlying generative replay for unsupervised learning during sleep or quiescence, and we propose that mechanistically this increase in variability is caused by the increased influence of top-down synapses that are not tied to incoming sensory stimuli.

Testable predictions

While our results are broadly consistent with existing experimental evidence, there are many unconfirmed aspects of our model which could be tested to validate or invalidate it (summarized in **Table 1** [\[48\]](#)). As mentioned in the previous section, our model predicts that *single neurons* should increase variability in response to psychedelic drug administration in any cortical area affected by psychedelic drugs. Second, we propose that psychedelic drugs should not push network dynamics into wildly different operating regimes than normal wakefulness, beyond any differences observed between wakefulness and replay (dream) during sleep. In particular, we found that our simulated psychedelic drug administration did not perturb pairwise correlations between neurons within local circuits when averaged across an ecologically representative set of stimuli.

Within our model, psychedelic drug administration is expected to increase the relative influence of top-down projections. This could be explored experimentally in several ways: first, we have shown that apical dendrite-targeted silencing experiments can identify the amount of influence apical dendritic inputs exert on neuronal dynamics; second, we have shown that increases in top-down influence can in principle be identified with interareal silencing experiments. We caution that interpreting results in this second vein may be difficult, as establishing a clean distinction between a ‘higher order’ and ‘lower order’ cortical area may be much more difficult in a densely recurrent system such as the brain, compared to our simplified and fully observable network model. Interestingly, if psychedelic drugs are genuinely exploiting circuitry ordinarily reserved for generative replay during periods of offline quiescence or sleep, we would expect that the same changes in functional connectivity observed during psychedelic drug administration would also occur during periods of replay. In the hippocampus, periods of replay have been tied to increases in apical dendritic influence [\[71\]](#), and increases in the strength of apical inputs have also been observed during NREM cortical updates in prefrontal cortex [\[49\]](#), though in this latter case it is unclear whether apical inputs affected the firing properties of the cell: thus, though there is some circumstantial evidence supporting the idea, much work remains to assess whether apical dendritic inputs dominate pyramidal neuron activity during replay or dreaming in cortex.

Though we have provided a candidate explanation for several of the effects of psychedelic drugs, our model rests on a number of testable assumptions. Our goal has been to articulate these assumptions as clearly as possible, to facilitate experimental efforts to test them.

Comparisons to alternative models

Aside from our model, there are two prominent existing hypotheses as to how psychedelic drugs could induce hallucinations in neural networks. The first proposed that incredibly complex, geometric patterns formed by DMT administration could be attributed to pattern-formation effects in visual cortex caused by a disruption of the balance between excitation and inhibition in locally-coupled topographic recurrent neural networks [\[72\]](#), [\[73\]](#). Our work differs from this approach in several respects. First, rather than disrupting E-I balance, we propose that psychedelics increase the relative influence of apical dendrites and top-down projections on the dynamics of neural activity. Second, though their model is able to generate geometric patterns, it is not able to generate patterns that are statistically related to the features of the sensory environment (e.g. MNIST digits). Lastly, for simplicity we avoided including topographic (or convolutional) recurrent connectivity in our model; however, it would be a very fruitful direction for future research to extend our work to generative modeling of temporal video sequences, as in [\[74\]](#), [\[75\]](#). With such a development, it is conceivable that our model could directly generalize these pattern formation-based approaches.

Testable Predictions
1. Psychedelic administration increases stimulus-conditioned variability of neurons.
2. Psychedelic administration preserves pairwise across-stimulus correlations between neurons.
3. Silencing apical dendritic compartments decreases neural variability more after psychedelic administration (and during replay events).
4. Silencing higher-order cortical areas affects lower-order cortical activity more after psychedelic administration.
5. Psychedelic drug effects are mediated by the same circuitry responsible for inducing generative replay dynamics in cortex.

Table 1

Summarizing testable predictions of the ‘oneirogen hypothesis’.

Perhaps more closely related to our model is the ‘relaxed beliefs under psychedelics’ (REBUS) model, which proposes to explain the effects of classical psychedelics in terms of predictive coding theory [60]. Similar to the Wake-Sleep algorithm, predictive coding theory [76] models sensory representation learning with neural dynamics and local synaptic modifications that collectively optimize an ELBO objective function. However, at a mechanistic level, there are numerous differences, the most easily distinguishable feature being that the Wake-Sleep algorithm *requires* periods of offline ‘generative replay’ to train bottom-up synapses in its network, whereas predictive coding learning occurs concomitantly with stimulus presentation. Furthermore, the REBUS model of psychedelic effects is described at a computational level, in terms of a decrease in the ‘precision-weighting of top-down priors.’ While it is more difficult to map the REBUS model directly onto cortical microcircuitry, and the hallucinatory effects of such a model have, to our knowledge, not been directly analyzed, it has been shown that the proposed mechanism causes an *increase* in bottom-up information flow between cortical areas [77], in direct contrast to the effects that we have shown in our model (**Figure 5c-d**). However, because genuinely causal interareal information flow can be difficult to analyze due to dense recurrent connectivity, we stress that it would be easier to distinguish between the REBUS model and our ‘oneirogen hypothesis’ by exploring whether psychedelic drugs affect the same circuitry that induces ‘generative replay’ during periods of sleep and quiescence.

Lastly, it should be noted that the Wake-Sleep algorithm and our choice of network architecture constitute one particular model within a family of related models, all of which satisfy our key criteria for a good model of the ‘oneirogen hypothesis,’ namely that 1) the model has well-defined top-down and bottom-up pathways, 2) it learns a generative model of incoming sensory inputs, and 3) it uses periods of offline replay for learning through local synaptic plasticity. For example, in the Supplemental Materials, we have replicated all of our essential results for two alternative network architectures, also learned via the Wake-Sleep algorithm: one model uses within-layer recurrence to improve generative performance, while the other model uses a simpler single compartment neuron model. Furthermore, the closely related Contrastive Divergence learning algorithm for Boltzmann Machines [78] also involves alternations between Wake and generative Sleep phases and learns through local synaptic plasticity, though Boltzmann machines are computationally more cumbersome to train and require more non-biological network features than the Wake-Sleep algorithm. We feel as though it is important to recognize that models that satisfy these three criteria are more similar than they are different, and that it may be quite difficult to experimentally distinguish between them.

Limitations

While our model is capable of capturing several effects of classical psychedelics, it also has several clear limitations. First, and most notably, our top-down generative model does not have sufficient expressive power to induce complex hallucinations of naturalistic stimuli (though it does a better job of modeling MNIST digits), producing instead ‘ripples,’ or ‘breathing’ effects. While psychedelic drug users do report these phenomena, they also report observing much more complex figures, including people, animals, and complex scenes [79, 80]. Generative models trained through backpropagation have been much more successful in producing more complex generated sensory stimuli [31, 30, 81], and even model hallucinations [82], but, as is well known, backpropagation is itself not a good model of learning in the brain [83]. This suggests that while it is quite possible for generative modeling approaches to produce complex hallucinations through non-biological means, algorithmic or architectural improvements may be necessary in order to make the performance of the more plausible Wake-Sleep algorithm closer to that achieved by backpropagation.

Our model also oversimplifies several aspects of biology. In particular, we do not use neurons that respect Dale’s law [84, 85], and the majority of our efforts to map the Wake-Sleep algorithm onto biology focus on excitatory pyramidal neurons. Furthermore, though we do observe that neural dynamics can tolerate a significant amount of top-down input before disrupting perception,

experiments and theoretical studies have shown that inputs to apical dendrites of pyramidal neurons do play an important role in waking perception [39, 86, 87], and are not *just* learning signals. We focused on clear distinctions between basally-driven Wake modes and apically-driven Sleep modes during training for computational efficiency reasons, and also due to the fact that parameter sharing across inference and generative networks in the Wake-Sleep algorithm is theoretically under-explored (though it is supported in closely related predictive coding approaches [76] and Boltzmann machines [78]).

Lastly, our modeling focus has been exclusively on cortical plasticity and hallucination effects: it should be noted that our model has little bearing on other important features of the psychedelic experience of potential therapeutic relevance, because we have not included the effects of psychedelics on subcortical structures including the serotonergic system [88], which plays an important role regulating mood and may be where psychedelics exert some of their antidepressant effects.

Conclusions

Here we have proposed a hypothesis for the mechanism of action of psychedelic drugs in terms of its excitatory effects on the apical dendrites of pyramidal neurons, which we propose pushes network dynamics into a state normally reserved for offline replay and learning; we have also proposed a number of testable predictions which could be used to validate or invalidate our hypothesis. If validated, our model would describe a mechanism by which the psychedelic experience causes ordinary sensory perception to become literally more dream-like; it further suggests that the plasticity increases observed during both sleep and psychedelic experience could occur via a common mechanism dedicated to sensory representation learning in the brain.

4 Methods

Model architecture and training

To model the effects of psychedelics on neural network dynamics and plasticity, we first constructed a simple model of the early visual system by training neural networks on two different image datasets (MNIST [53] and CIFAR10 [54]). Networks were trained with the Wake-Sleep algorithm [29], which requires, for each layer, two modes of stochastic network activity: a ‘generative mode’, and an ‘inference mode’. For the ‘inference’ mode we must specify a probability distribution $b(\mathbf{r}^{(l)} \mid \mathbf{r}^{(l-1)})$, while for the ‘generative’ mode we must specify a separate distribution $p(\mathbf{r}^{(l)} \mid \mathbf{r}^{(l+1)})$. As a notational convention, we will use letters when referring to mathematical objects from the generative, top-down distribution, and their vertical reflection when referring to the inference, bottom-up distribution (e.g. p and b). Notice here that activity in ‘inference’ mode is conditioned on ‘bottom-up’ network states ($\mathbf{r}^{(l-1)}$), while activity in generative mode is conditioned on ‘top-down’ network states ($\mathbf{r}^{(l+1)}$) (Figure 1a).

The ‘inference mode’ specifies a probability distribution over neural activity, conditioned on the next-lower layer (where the lowest layer is the stimulus layer, i.e. $\mathbf{r}^{(0)} = \mathbf{s}$)—mechanistically it corresponds to activity generated by feedforward projections. To increase the expressive power of our neural units, we use multicompartmental neuron models similar to [89] with N_d dendritic compartments, whose voltages are summed nonlinearly to form the full input to the basal dendrites. For $l > 0$, layer activity is sampled from the distribution [ineq, where for neuron i in layer l , $h_i(\mathbf{r}^{(l-1)})$ is given by:

$$h_i(\mathbf{r}^{(l-1)}) = \phi \left(\sum_{n=0}^{N_d} w_{in}^{(l)} \phi_d \left(\mathbf{w}_{in}^{(l)} \mathbf{r}^{(l-1)} + c_{in}^{(l)} \right) + b_i^{(l)} \right), \quad (2)$$

where $\mathbf{w}_{in}^{(l)}$ is a $1 \times N^{(l-1)}$ matrix of synaptic weights onto dendrite n , $c_{in}^{(l)}$ is the corresponding bias for the n th dendritic compartment, $w_{in}^{(l)}$ is the strictly positive weight given to the n th dendritic branch (roughly corresponding to a conductance), and $b_i^{(l)}$ is the bias for the entire basal compartment. $\Phi_d(\cdot)$ and $\Phi(\cdot)$ are nonlinearities for the dendritic branches and the total basal compartment, respectively: both are the sequential composition of the tanh nonlinearity, followed by batch normalization [90]. For the dendritic branch nonlinearities, we allow for learnable affine parameters (scale and bias), but for the entire basal dendritic compartment we constrain activity to be zero-mean and unit variance across batches in order to prevent indeterminacy between apical and basal scale parameters. For the final inference layer $\mathbf{r}^{(L)}$, as in the variational autoencoder [30], we parameterize both the mean and a diagonal covariance matrix of the inference distribution: $\mathbf{r}^{(L)} \sim \mathcal{N}(h(\mathbf{r}^{(L-1)}), \text{diag}(h_2(\mathbf{r}^{(L-1)})))$, where $h_2(\cdot)$ is also a multicompartmental model, in this case replacing the final batch normalization with an exponential nonlinearity to ensure positivity.

The ‘generative’ mode specifies a probability distribution over neural activity, conditioned on the next-higher layer—it corresponds mechanistically to activity generated by feedback projections. The highest layer, $\mathbf{r}^{(L)}$ is sampled from an $N^{(L)}$ -dimensional independent standard normal distribution, $\mathbf{r}^{(L)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, and all subsequent layers are sampled from the distribution $\mathbf{r}^{(l)} \sim \mathcal{N}(\mu(\mathbf{r}^{(l+1)}), \sigma_p^2 \mathbf{I})$, where for the i th neuron, $\mu_i(\mathbf{r}^{(l+1)})$ is given by:

$$\mu_i(\mathbf{r}^{(l+1)}) = \phi \left(\sum_{n=0}^{N_d} m_{in}^{(l)} \phi_d \left(\mathbf{M}_{in}^{(l)} \mathbf{r}^{(l+1)} + d_{in}^{(l)} \right) + a_i^{(l)} \right), \quad (3)$$

where $\mathbf{M}_{in}^{(l)}$ is a $1 \times N^{(l+1)}$ matrix of synaptic weights onto apical dendritic branch n , $d_{in}^{(l)}$ is the corresponding bias for the n th dendritic compartment, $m_{in}^{(l)}$ is the strictly positive weight given to the n th dendritic branch, and $a_i^{(l)}$ is the bias for the entire apical compartment. Again, $\Phi_d(\cdot)$ and $\Phi(\cdot)$ are nonlinearities, identical to the inference (basal) pathway.

While the neuron model used here is more complicated than is normally used for single-unit neuron models, functions of this kind could feasibly be implemented by nonlinear dendritic computations [89]; we further found that using this nonlinearity qualitatively improved generative performance (Supplemental Figure S2). Given these parameterized probability distributions, we then determined the neural activity for each layer l according to Eq. 1. Our network trained on MNIST was composed of 3 layers, with widths [32, 16, 6], listed in ascending order. A full list of network hyperparameters for both our MNIST and CIFAR10-trained networks can be found in the Supplemental Methods.

All synaptic weights and parameters in our networks were trained via the Wake-Sleep algorithm [29], which is known to produce ‘local’ parameter updates for a wide range of neuron models (and rate or spike-based output distributions), though the specific functional form of the update may vary depending on the neuron model chosen [91]. These updates, for reasonable choices of neural network architecture, can be interpreted as predictions for how synaptic plasticity should look in the brain, if learning were really occurring via the Wake-Sleep algorithm or some approximation thereof.

Consider a generic inference (basal dendrite) parameter for neuron i , $i, \theta_b \in \{w_{in}^{(l)}, \mathbf{w}_{in}^{(l)}, b_i^{(l)}, c_{in}^{(l)} : n = 0, \dots, N_d\}$. The Wake-Sleep algorithm gives the following update, for a single stimulus presentation:

$$\Delta \theta_b = (\alpha) \eta \frac{\left(\mathbf{r}_i^{(l)} - h_i(\mathbf{r}^{(l-1)}, \theta_b) \right)}{\sigma_b^2} \frac{\partial h_i(\mathbf{r}^{(l-1)}, \theta_b)}{\partial \theta_b}, \quad (4)$$

where η is a learning rate, and the gate α ensures that learning only occurs during sleep mode. Further, for reasons of computational efficiency, we average weight updates across a batch of 512 stimulus presentations; similar results could in principal be obtained with purely online updates [92], but we opted to present stimuli in batches here in order to parallelize computations. $\frac{\partial h_i(\mathbf{r}^{(l-1)}, \theta_b)}{\partial \theta_b}$ changes depending on the parameter θ , reflecting that particular parameter's contribution to basal dendritic activity. For a dendritic branch weight $w_{in}^{(l)}$ we have:

$$\frac{\partial h_i(\mathbf{r}^{(l-1)}, w_{in}^{(l)})}{\partial w_{in}^{(l)}} = \phi'(\mathbf{v}_i^{total}) \phi_d(\mathbf{v}_{in}), \quad (5)$$

where $\mathbf{v}_i^{total}$ is the total input to the basal dendritic compartment, and $\mathbf{v}_{in} = \mathbf{W}_{in}^{(l)} \mathbf{r}^{(l-1)} + c_{in}^{(l)}$ is the total input to the n th dendritic branch. This update has the functional form of a classical 'delta' learning rule [65], where a compartmental prediction error between local dendritic activity and neuronal firing rate is multiplicatively combined with branch-specific input to provide changes in the conductance for the n th branch. Similarly, for the j th synapse on the n th dendritic branch, $\mathbf{W}_{inj}^{(l)}$, we have:

$$\frac{\partial h_i(\mathbf{r}^{(l-1)}, \mathbf{W}_{inj}^{(l)})}{\partial \mathbf{W}_{inj}^{(l)}} = \phi'(\mathbf{v}_i^{total}) w_{in}^{(l)} \phi'(\mathbf{v}_{in}) \mathbf{r}_j^{(l-1)}. \quad (6)$$

Unlike for simple one-compartment neuron models, computation of parameter updates for dendritic synapses $\mathbf{W}_{inj}^{(l)}$ requires weighting the 'delta' error by the conductance of the corresponding dendritic branch (w_{in}), which could be approximated by the passive diffusion of signaling molecules from the principal basal dendritic compartment back along dendritic branches to individual synapses.

For generative parameters ($\theta_p \in \{m_{in}^{(l)}, \mathbf{M}_{in}^{(l)}, a_i^{(l)}, d_{in}^{(l)} : n = 0, \dots, N_d\}$), we have a nearly identical update for a single stimulus presentation:

$$\Delta \theta_p = (1 - \alpha) \eta \frac{(\mathbf{r}_i^{(l)} - \mu_i(\mathbf{r}^{(l+1)}, \theta_p))}{\sigma_p^2} \frac{\partial \mu_i(\mathbf{r}^{(l+1)}, \theta_p)}{\partial \theta_p}, \quad (7)$$

where now input in the apical dendritic compartment, $\mu_i(\mathbf{r}^{(l+1)})$, is being compared to the activity of the neuron as a whole to determine the magnitude and sign of plasticity. The $(1 - \alpha)$ gate in this case ensures that plasticity only occurs during the Wake mode.

We provide pseudocode (Algorithm 1) for our Wake-Sleep implementation, as well as a full list of algorithm and optimizer hyperparameters (Tables S1 and S2) in the Supplemental Materials. Code for reproducing all results from this study is available here: <https://github.com/colinbrendenberg/oneirogen-hypothesis>.

Modeling Hallucinations

During training, neural network activity is either dominated entirely by bottom-up inputs (Wake, $\alpha = 0$) or by top-down inputs (Sleep, $\alpha = 1$). As a consequence, sampling neural activity is computationally low-cost, and can be performed in a single time step. During Wake, one can take a sampled stimulus variable $\mathbf{s}$, determine the activity at layer 1, then 2, and so on until layer L , while during Sleep, one can sample a latent network state in layer L and traverse the layers in reverse

order, down to the stimulus layer. However, this is not possible if $\alpha \notin \{0, 1\}$, because activity in each layer l should depend simultaneously on layer $l + 1$ and layer $l - 1$. For this reason, we chose to model hallucinatory neural activity *dynamically*, as follows:

$$\mathbf{r}_t^{(l)} = (1 - \frac{1}{\tau})\mathbf{r}_{t-1}^{(l)} + \frac{1}{\tau}f(h(\mathbf{r}_{t-1}), \mu(\mathbf{r}_{t-1}), \alpha) + \frac{f(\sigma_b, \sigma_p, \alpha)}{\sqrt{\tau}}\boldsymbol{\eta}_{t-1}, \quad (8)$$

where τ is a time constant that determines how much of the previous network state is retained, and $\boldsymbol{\eta}_{t-1} \sim \mathcal{N}(0, \mathbf{I})$. Critically, if we take $\tau = 1$ these dynamics reduce to the sampling procedure used during training (Eq. 1). *A priori*, the choice of interpolation function $f(a, b, \alpha)$ is arbitrary. We selected the following function:

$$f(a, b, \alpha) = \kappa \log \left[(1 - \alpha) \exp \frac{a}{\kappa} + \alpha \exp \frac{b}{\kappa} \right], \quad (9)$$

where $\kappa = 0.35$ is a free parameter. This function is equivalent to linear interpolation as $\kappa \rightarrow \infty$, and is equivalent to the maximum function between arguments a and b as $\kappa \rightarrow 0$ if $\alpha = 0.5$. By selecting $\kappa = 0.35$, we are biasing the system towards registering positive inputs from apical or basal sources (in the inclusive sense). We found that this produced ‘hallucinatory’ percepts in stimulus space that did not reduce the intensity of input stimuli as α increased; rather, inputs maintained their intensity and hallucinations were added on top if they were of greater intensity than the ground-truth image. All simulations were run for 800 timesteps, with $\tau = 0.1$. As a control, we compared our results to network dynamics produced purely by increases in noise, without increases in apical dendritic influence (which we refer to as our noise-based hallucination protocol). For these control simulations, we produced network activity time series with the following equation:

$$\mathbf{r}_t^{(l)} = (1 - \frac{1}{\tau})\mathbf{r}_{t-1}^{(l)} + \frac{1}{\tau}h(\mathbf{r}_{t-1}) + \frac{\sigma_b + \alpha}{\sqrt{\tau}}\boldsymbol{\eta}_{t-1}, \quad (10)$$

so that the standard deviation of the injected noise increased linearly with α .

Apical and Basal Alignment

To measure the alignment between inputs in the apical and basal dendritic compartments of our model neurons, we computed the ‘Wake’ neural responses to the full test dataset and measured the activity in both the basal and apical compartments of our neurons ($h(\mathbf{r}^{(l-1)})$ and $\mu(\mathbf{r}^{(l+1)})$, respectively). We then calculated the correlation coefficient between apical and basal compartments for the same neuron, compared to the correlation between compartments for two randomly selected neurons.

Quantifying plasticity

To quantify the total amount of plasticity induced in our model system by the administration of psychedelic drugs, we measured the change in *relative* parameter strength (averaging across all synapses in the network and an ensemble of 512 test images). For each test image, we simulated network dynamics according to Eq. 8. Subsequently, for each parameter θ , we calculated the net amount of plasticity induced by viewing all test images, $\Delta\theta$. We subsequently reported the relative change:

$$\Delta\theta_{rel} = \frac{|\Delta\theta|}{|\theta| + \epsilon}, \quad (11)$$

under conditions in which α values gate plasticity (as in ordinary Wake-Sleep) and under conditions in which psychedelic drug administration does not also affect plasticity gating. Here, we took $\epsilon = 10^{-2}$ to avoid numerical instabilities.

Classifier training

As we trained our neural network using the Wake-Sleep algorithm, we simultaneously trained a *separate* classifier network based on Wake-phase neural activity in the second network layer on a cross-entropy loss, to identify the stimulus class of the input to the system. For our classifier, we used a multilayer perceptron neural network with a single 256-unit hidden layer and $\tanh(\cdot)$ nonlinearities.

We then quantified the accuracy of the classifier on the test set, based on neural activity drawn from the final time step T of hallucination simulations with various values of α . We further measured the average variance of the 10-dimensional output logits of the neural network.

Quantifying correlation matrix similarity before and after psychedelics

To quantify how similar the pairwise correlations between neurons in our model networks were before and after the administration of psychedelics, we recorded hallucinatory network dynamics for an ensemble of 512 test images, and measured pairwise correlations between neurons in the first network layer. To compare these matrices, we then report the correlation coefficient between the flattened $N \times N$ matrices. For this metric, a value of 1 indicates that the correlation matrices are perfectly aligned, while a value of -1 indicates that pairwise correlations are fully inverted.

Quantifying interareal causality through inactivations

To quantify changes in interareal functional connectivity induced by psychedelics, we performed two different types of inactivation. In the first, we inactivated the apical dendritic compartments of all neurons in the stimulus layer, and measured how this inactivation affected across-stimulus variability of neurons relative to the fully active state. In the second method, we inactivated all neurons in the deepest layer, and measured the same effect in across-stimulus variability in the stimulus layer. For both inactivation schemes, we report the mean and standard error of the variance ratio:

$$VR = \frac{\text{Var}_{inact}(\mathbf{r}^{(0)}) + \epsilon_v}{\text{Var}(\mathbf{r}^{(0)}) + \epsilon_v}, \quad (12)$$

where we added $\epsilon_v = 10^{-3}$ to the denominator to prevent numerical instability and to the numerator ensure that the ratio evaluates to 1 if the two variances are equivalent.

6 Ethics Declarations

Psychedelic drug research has a long history fraught with many instances of unethical research practice [93]. Further, psychedelic drug use itself has long been stigmatized and punished through legal measures [94], often at the expense of indigenous peoples who have incorporated psychoactive substances into their cultural and spiritual practices for millennia [5]. In the interest of avoiding a repetition of past mistakes, we feel compelled to provide explicit guidance on how our work should be interpreted and used. To do so, we will take inspiration from two principal ethical frameworks: the Montreal Declaration on Responsible AI [95], and the EQUIP framework for equity-oriented healthcare [96, 97]. We strongly encourage anyone considering extending our research or using our work in any form of clinical setting to ensure that subsequent research adheres to these frameworks.

Below, drawing from these ethical frameworks, we will provide a set of guidelines for how our work should be interpreted and used. Though these guidelines are by no means exhaustive, our hope is that adherence to them will help promote the potential positive outcomes of our work while limiting potential negative consequences.

Guidelines for the ethical use of this study

Do:

1. Ensure that the elements of our hypothesis have been adequately tested, as outlined in our discussion, *before* using our framework in any form of clinical or therapeutic setting.
2. Use our ideas to inform further basic neuroscience research on perception, learning, sleep, and replay phenomena.
3. Explore our ideas as an opportunity to inform your own understanding of cognition, learning, and perception, with the understanding that these ideas have not yet been validated experimentally.
4. Feel free to ask us if you are worried that your proposed use of our work may have negative impacts.

Do not:

1. Report our results as scientific fact. We have outlined a *hypothesis*, which is designed to be tested by the experimental neuroscience community.
2. Cite or interpret our results without an adequate understanding of the mathematics involved. Feel free to ask us if you are worried that you may be misinterpreting our results.
3. Use our results to extract undue or inequitable profit. The ideas developed in this paper are the product of decades of research and public funding, built upon *millennia* of exploration of psychedelics. Any knowledge or value contained within this paper is the common heritage of all humanity, with particular recognition due to the indigenous and marginalized communities that have historically suffered and are currently suffering from oppressive government and industry policies.
4. Use our results for any application that could violate human rights or harm human beings in any way.

A Supplementary Materials

Recurrent network model

To explore the extent to which our results hold for different neuron models, and to give our generative model more expressive power than the traditional Helmholtz machine [98], we constructed a network model with a single timestep of within-layer recurrent denoising in each layer, which gives our model some similarities to denoising diffusion approaches [99]. For both our ‘inference’ mode and our ‘generative’ mode we specify both a denoised network state $\bar{\mathbf{r}}^{(l)}$ and a noise-corrupted network state $\mathbf{r}^{(l)}$ for layer l ; specifying a neural network model is then equivalent to specifying, for each layer, a joint probability distribution over denoised and noise-corrupted network states for both the inference and generative modes, i.e. for the ‘inference’ mode we must specify a probability distribution $b(\bar{\mathbf{r}}^{(l)}, \mathbf{r}^{(l)} | \bar{\mathbf{r}}^{(l-1)})$, while for the ‘generative’ mode we must specify a separate distribution $p(\bar{\mathbf{r}}^{(l)}, \mathbf{r}^{(l)} | \bar{\mathbf{r}}^{(l+1)})$. As a notational convention, we will use letters when referring to mathematical objects from the generative, top-down distribution, and their vertical reflection when referring to the inference, bottom-up distribution (e.g. p and b).

Notice here that activity in ‘inference’ mode is conditioned on ‘bottom-up’ network states ($\mathbf{r}^{(l-1)}$), while activity in generative mode is conditioned on ‘top-down’ network states ($\bar{\mathbf{r}}^{(l+1)}$) (**Figure 1a** [↗](#)).

The ‘inference mode’ specifies a probability distribution over neural activity, conditioned on the next-lower layer (where the lowest layer is the stimulus layer, i.e. $\mathbf{r}^{(0)} = \mathbf{s}$)—mechanistically it corresponds to activity generated by feedforward projections. For $l > 0$, layer activity is sampled from the distribution $\mathbf{r}^{(l)} \sim \mathcal{N}(h(\mathbf{r}^{(l-1)}), \sigma_b^2 \mathbf{I})$, where $h(\mathbf{r}^{(l-1)})$ is given by:

$$h(\mathbf{r}^{(l-1)}) = \tanh(\mathbf{W}^{(l)} \mathbf{r}^{(l-1)} + b^{(l)}). \quad (13)$$

Subsequently, we add additional noise to get a noise-corrupted network state $\bar{\mathbf{r}}^{(l)} \sim \mathcal{N}((1 - \bar{\sigma}_b^2) \mathbf{r}^{(l)}, \bar{\sigma}_b^2 \mathbf{I})$; while noise corruption is a natural feature of network dynamics in the brain [100 [↗](#)], we include it here in our model because it has been shown that *denoising* is a critical aspect of many powerful generative modeling approaches [101 [↗](#), 102 [↗](#), 103 [↗](#)], and we have likewise found that it improves the quality of generated images in our learned networks (**Supplemental Figure S2** [↗](#)).

The ‘generative’ mode specifies a probability distribution over neural activity, conditioned on the next-higher layer—it corresponds mechanistically to activity generated by feedback projections. The highest layer, $\mathbf{r}^{(L)}$ is sampled from an $N^{(L)}$ -dimensional independent standard normal distribution, $\mathbf{r}^{(L)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, and all subsequent layers are sampled from the distribution $\mathbf{r}^{(l)} \sim \mathcal{N}(\mu(\bar{\mathbf{r}}^{(l+1)}), \sigma_p^2 \mathbf{I})$, where $\mu(\bar{\mathbf{r}}^{(l+1)})$ is given by:

$$\mu(\bar{\mathbf{r}}^{(l+1)}) = \tanh(\mathbf{M}^{(l)} \bar{\mathbf{r}}^{(l+1)} + a^{(l)}), \quad (14)$$

where $m^{(l)}$ is a $N^l \times N^{(l+1)}$ weight matrix, and a is a bias term. Subsequently, the network goes through a single timestep of recurrent denoising, so that $\bar{\mathbf{r}}^{(l)} \sim \mathcal{N}(\bar{\mu}(\mathbf{r}^{(l)}), \bar{\sigma}_p^2 \mathbf{I})$, where $\bar{\mu}(\mathbf{r}^{(l)})$ is given by:

$$\bar{\mu}(\mathbf{r}^{(l)}) = \mathbf{r}^{(l)} + \sigma(\mathbf{C}_1 \mathbf{r}^{(l)} + \mathbf{c}_1) \tanh(\mathbf{C}_2 \mathbf{r}^{(l)} + \mathbf{c}_2), \quad (15)$$

where $\sigma(\cdot)$ is a sigmoid nonlinearity that acts as a gating function similar those used in the LSTM [104 [↗](#)] and GRU [105 [↗](#)], $\mathbf{C}_1$ and $\mathbf{C}_2$ are $N^{(l)} \times N^{(l)}$ recurrent weight matrices, and $\mathbf{c}_1$ and $\mathbf{c}_2$ are biases. While this is a more complicated nonlinearity than is normally used for single-unit neuron models, functions of this kind could feasibly be implemented by nonlinear dendritic computations [89 [↗](#)]; we further found that using this nonlinearity qualitatively improved generative performance. Given these parameterized probability distributions, we then determined the neural activity for each layer l according to Eq. (1) [↗](#). As with our multicompartmental neuron model, inference and generative parameters were updated according to Eqs. (4) [↗](#) and (7) [↗](#), respectively. Recurrent network hyperparameters are available in **Table S3** [↗](#).

Simplified neuron model

As a control, we also tested our results using a simplified multilayer perceptron neuron model, which used neither batch normalization nor multiple dendritic branches. For the ‘inference’ mode within the simplified model, for $l > 0$, layer activity is sampled from the distribution $\mathbf{r}^{(l)} \sim \mathcal{N}(h(\mathbf{r}^{(l-1)}), \sigma_b^2 \mathbf{I})$, where for neuron i in layer l , $h_i(\mathbf{r}^{(l-1)})$ is given by:

$$h_i(\mathbf{r}^{(l-1)}) = \tanh(\mathbf{W}_{i:}^{(l)} \mathbf{r}^{(l-1)} + b_i^{(l)}), \quad (16)$$

where $\mathbf{w}_{i:}^{(l)}$ is a $1 \times N^{(l-1)}$ matrix of basal synaptic weights onto neuron i , and b_i is the corresponding bias.

The simplified ‘generative’ mode likewise replaces the branched neuron model used in the main text with a multilayer perceptron model. The highest layer, $\mathbf{r}^{(L)}$ is sampled from an $N^{(L)}$ -dimensional independent standard normal distribution, $\mathbf{r}^{(L)} \sim \mathcal{N}(0, \mathbf{I})$, and all subsequent layers are sampled from the distribution $\mathbf{r}^{(l)} \sim \mathcal{N}(\mu(\mathbf{r}^{(l+1)}), \sigma_p^2 \mathbf{I})$, where for the i th neuron, $\mu_i(\mathbf{r}^{(l+1)})$ is given by:

$$\mu_i(\mathbf{r}^{(l+1)}) = \tanh \left(\mathbf{M}_{i:}^{(l)} \mathbf{r}^{(l+1)} + a_i^{(l)} \right), \quad (17)$$

where $\mathbf{M}_{i:}^{(l)}$ is a $1 \times N^{(l+1)}$ matrix of apical synaptic weights onto neuron i , and $a_i^{(l)}$ is the corresponding bias. As with the branched neuron model, inference and generative parameters were updated according to **Equations (4)** and **(7)**, respectively. For optimization, we used the identical hyperparameters to the multicompartment neuron model (**Table S1**).

Figure S1

Visualizing the effects of psychedelics for alternative model architectures.

We model the effects of classical psychedelics by progressively increasing α from 0 to 1 in alternative model architectures. We visualize the effects of psychedelics on the network representation by inspecting the stimulus layer **s**. **a)** Example stimulus-layer activity (rows) in response to an MNIST digit presentation as psychedelic dose increases (columns, left to right) in the recurrent network model. **b)** Same as (a) but for our single compartment neuron model. **c)** Same as (a) using the multicompartment neuron model used for our main results, but for our noise-based hallucination protocol. **d)** Same as (c), but in a network in which neither the generative nor inference pathways have been trained beyond initialization.

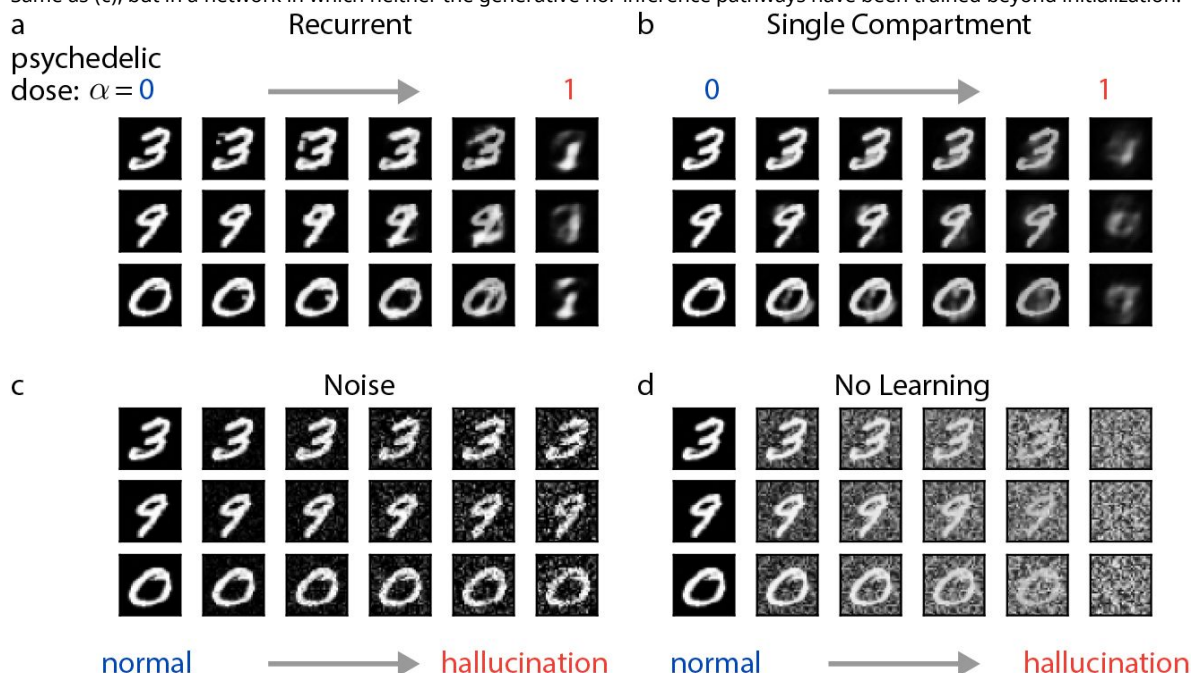


Figure S2

Example generated images for different model architectures and datasets.

Generated images sampled from Eq. (1) with $\alpha = 1$ for: **a)** Our primary multicompartment neuron model trained on MNIST, **b)** A multicompartment neuron model trained on CIFAR10, **c)** The recurrent network model, **d)** The single compartment neuron model.

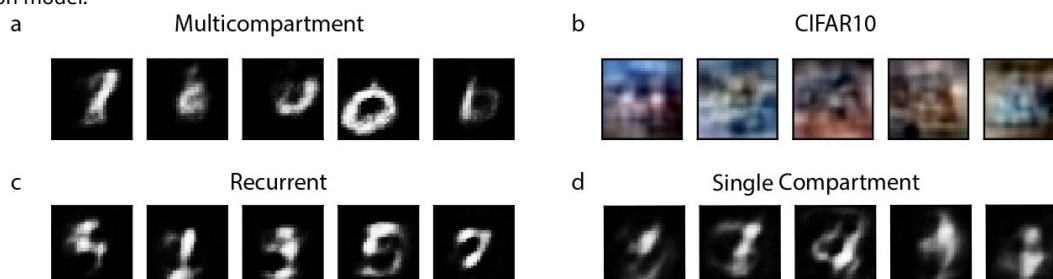


Figure S3

Alignment between apical and basal dendritic compartments for different model architectures and datasets.

Apical-basal alignment for: **a)** An untrained multicompartment neuron model trained on MNIST, **b)** A single compartment neuron model, **c)** A recurrent network model, **d)** A multicompartment neuron model trained on CIFAR10.

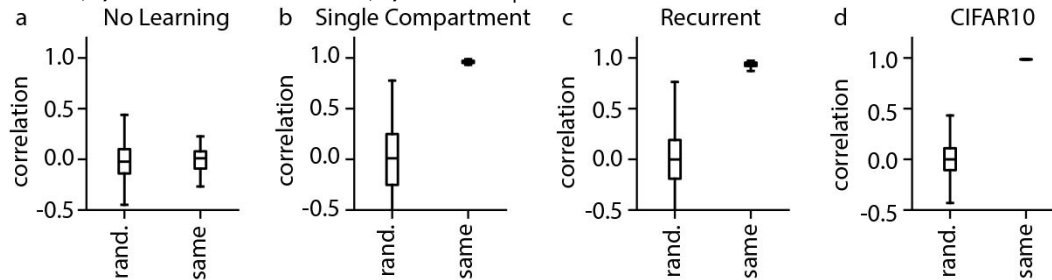
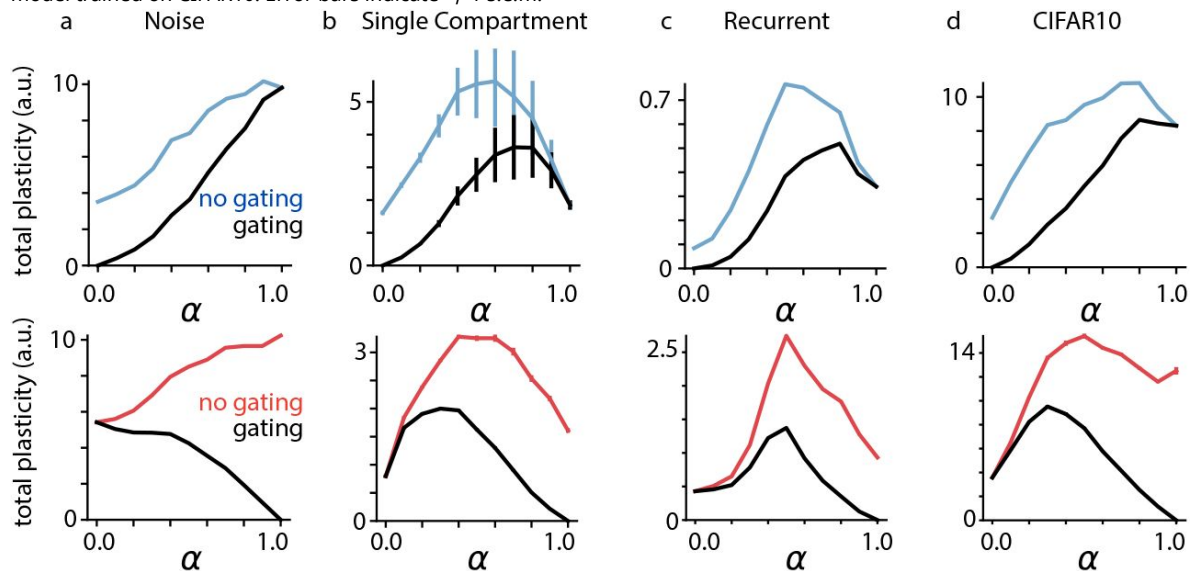


Figure S4

Hallucination-induced synaptic plasticity for different neuron models.

a) Basal (top) and apical (bottom) plasticity as a function of α for a multicompartment neuron model trained on MNIST, using our noise-based hallucination protocol as a control. **b)** Same as (a) for a single compartment neuron model, using our primary hallucination protocol. **c)** Same as (b) for a recurrent network model, **d)** Same as (b) for a multicompartment neuron model trained on CIFAR10. Error bars indicate ± 1 s.e.m.



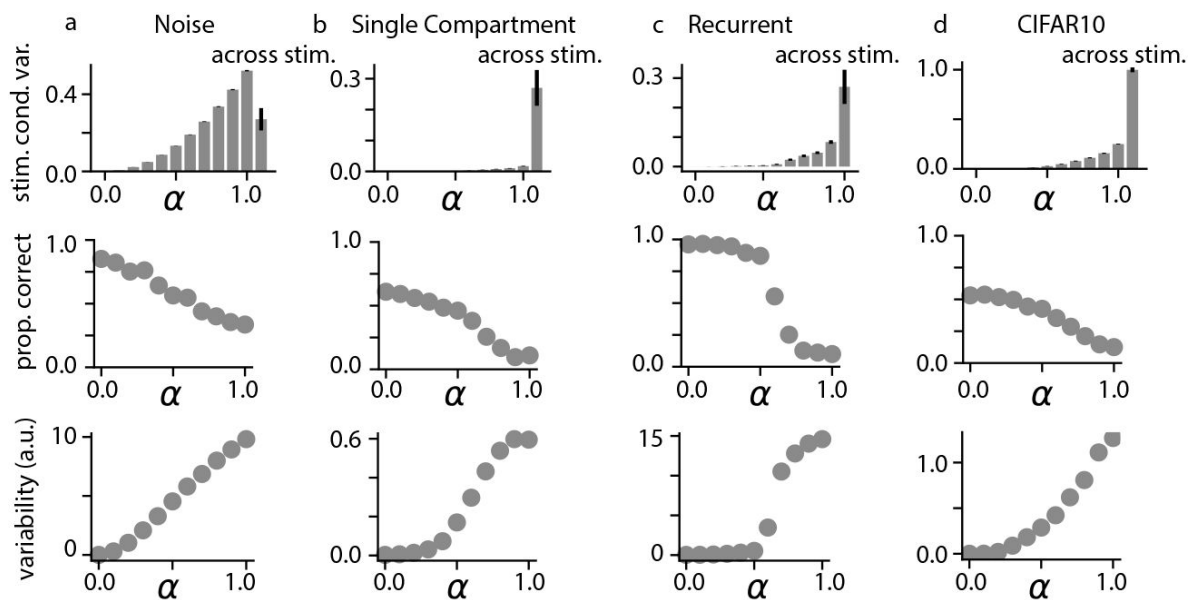


Figure S5

Neural variability changes for different neuron models.

a) Stimulus-conditioned variability (top), classifier accuracy (middle), and classifier output variability (bottom) as a function of α for a multicompartiment neuron model trained on MNIST, using our noise-based hallucination protocol as a control. **b)** Same as (b) for a single compartment neuron model, using our primary hallucination protocol. **c)** Same as (b) for a recurrent network model, **d)** Same as (b) for a multicompartiment neuron model trained on CIFAR10. Error bars indicate ± 1 s.e.m.

Figure S6

Network-level effects of psychedelics for different network architectures and training datasets.

For each network architecture, we examine: correlation similarity as a function of α (top row), the proportion explained variance across stimuli as a function of principal component number (second row), the ratio of across-stimulus variance in stimulus layer neurons when apical dendrites have been inactivated compared to baseline conditions across different α values (third row), and the ratio of across-stimulus variance in stimulus layer neurons when the deepest network layer has been inactivated across different α values (fourth row). **a)** Results for an untrained multicompartment neuron. **b)** Results for a multicompartment neuron model trained on MNIST, using our noise-based hallucination protocol. **c)** Results for a single compartment neuron model. **d)** Results for a recurrent network model. **e)** Results for a multicompartment neuron model trained on CIFAR10. Error bars indicate ± 1 s.e.m.

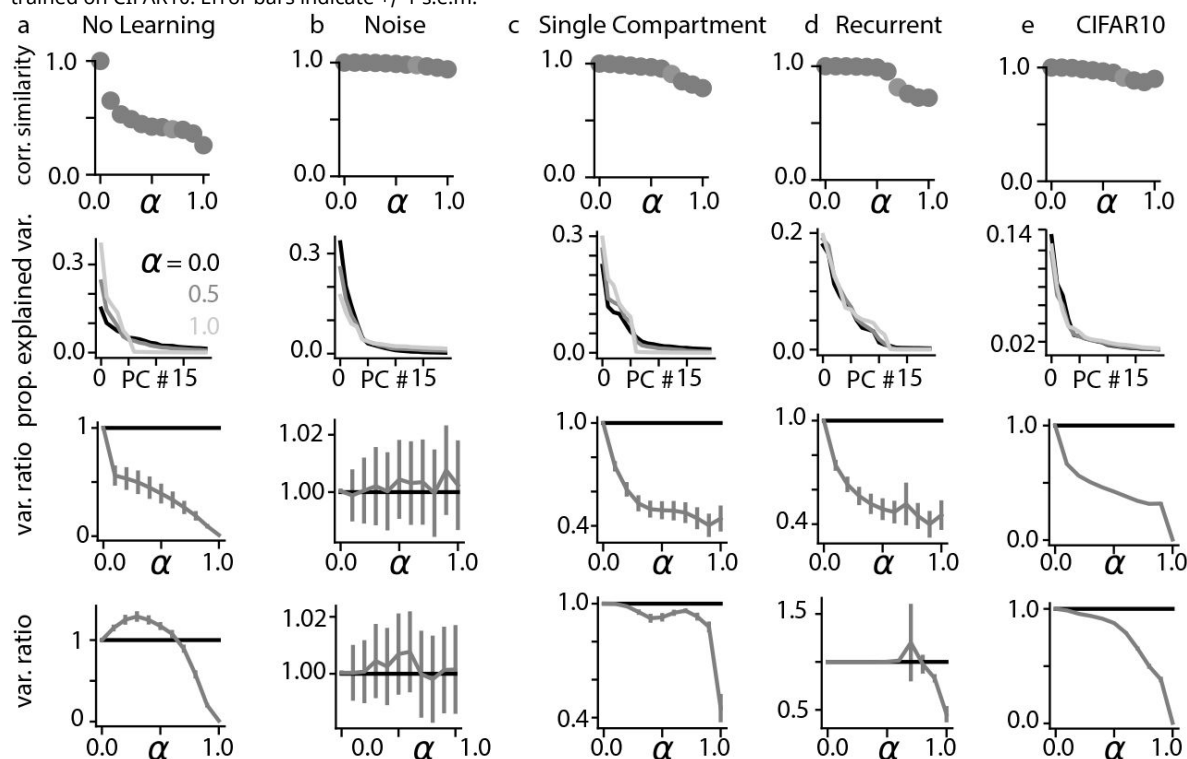


Table S1

MNIST multicompartment network hyperparameters

Parameter	Function	Value
$\{N^{(l)}\}$	Layer Widths	[32, 16, 6]
σ_b	Inference standard dev.	0.01
σ_p	Generative standard dev.	0.01
N_d	Number of dendritic branches	16
L	Epoch number	50
T_w	Length of Wake phase	200
T_s	Length of Sleep phase	200
K	Batch number	512
η_p	Generative parameter learning rate	0.001
η_b	Inference parameter learning rate	0.001
$[\beta_1, \beta_2]$	Adam optimizer parameters	[0.9, 0.999]

Table S2

CIFAR10 multicompartment network hyperparameters

Parameter	Function	Value
$\{N^{(l)}\}$	Layer Widths	[128, 128, 96]
σ_b	Inference standard dev.	0.01
σ_p	Generative standard dev.	0.01
N_d	Number of dendritic branches	16
L	Epoch number	50
T_w	Length of Wake phase	200
T_s	Length of Sleep phase	200
K	Batch number	256
η_p	Generative parameter learning rate	0.001
η_b	Inference parameter learning rate	0.001
$[\beta_1, \beta_2]$	Adam optimizer parameters	[0.9, 0.999]

Table S3

Recurrent network hyperparameters

Parameter	Function	Value
$\{N^{(l)}\}$	Layer Widths	[512, 256, 256, 128, 64, 32, 16]
σ_b	Denoised inference standard dev.	0.01
$\bar{\sigma}_b$	Noisy inference standard dev.	0.01
σ_p	Denoised generative standard dev.	0.01
$\bar{\sigma}_p$	Noisy generative standard dev.	0.01
L	Epoch number	400
T_w	Length of Wake phase	200
T_s	Length of Sleep phase	50
K	Batch number	512
η_p	Generative parameter learning rate	0.001
η_b	Inference parameter learning rate	0.0001
$[\beta_1, \beta_2]$	Adam optimizer parameters	[0.9, 0.999]

Algorithm 1 Wake-Sleep Pseudocode Algorithm 1 Wake-Sleep Pseudocode

```

1:  $e \leftarrow 0$ 
2:  $\mathcal{L}_{prev} \leftarrow \infty$ 
3: while  $e < L$  do ▷ Repeat for  $L$  data epochs
4:   for  $t$  in range( $0, T_w$ ) do ▷ Wake Phase
5:     for  $k$  in range( $0, K$ ) do ▷ Sample network states
6:        $\mathbf{r}^{(k)} \sim b(\mathbf{r}, \theta_b)$ 
7:       if epoch is complete then  $e \leftarrow e + 1$ 
8:       end if
9:     end for
10:     $\mathcal{L}(\theta_p) \leftarrow \frac{1}{K} \sum_{k=0}^K -\log p(\mathbf{r}^{(k)}, \theta_p)$  ▷ Update generative parameters  $\theta_p$ 
11:     $\Delta\theta_p \leftarrow \eta_p \nabla_{\theta_p} \mathcal{L}(\theta_p)$ 
12:     $\theta_p \leftarrow \text{Adam}(\theta_p, \Delta\theta_p)$ 
13:  end for
14:  if  $\mathcal{L}(\theta_p) < \mathcal{L}_{prev}$  then
15:     $\mathcal{L}_{prev} \leftarrow \mathcal{L}(\theta_p)$ 
16:    for  $t$  in range( $0, T_s$ ) do ▷ Sleep Phase
17:      for  $k$  in range( $0, K$ ) do ▷ Sample network states
18:         $\mathbf{r}^{(k)} \sim p(\mathbf{r}, \theta_p)$ 
19:      end for
20:       $\mathcal{L}(\theta_b) \leftarrow \frac{1}{K} \sum_{k=0}^K -\log b(\mathbf{r}^{(k)}, \theta_b)$  ▷ Update inference parameters  $\theta_b$ 
21:       $\Delta\theta_b \leftarrow \eta_b \nabla_{\theta_b} \mathcal{L}(\theta_b)$ 
22:       $\theta_b \leftarrow \text{Adam}(\theta_b, \Delta\theta_b)$ 
23:    end for
24:  end if
25: end while

```

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Additional files

Figure 2a multicompartmental mnist hallucination video [🔗](#)

Figure 2c multicompartmental cifar10 hallucination video [🔗](#)

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Reviewer #1 (Public review):

Bredenberg et al. aim to model some of the visual and neural effects of psychedelics via the Wake-Sleep algorithm. This is an interesting study with findings that go against certain mainstream ideas in psychedelic neuroscience (that I largely agree with). I cannot speak to the math in this manuscript, but it seems like quite a conceptual leap to set a parameter of the model in between wake and sleep and state that this is a proxy to acute psychedelic effects (point #20). My other concerns below are related to the review of the psychedelic literature:

(1) Page 1, Introduction, "...they are agonists for the 5-HT_{2A} serotonin receptor commonly expressed on the apical dendrites of cortical pyramidal neurons..." It is a bit redundant to say "5-HT_{2A} serotonin receptor," as serotonin is already captured by its abbreviation (i.e., 5-HT).

While psychedelic research has focused on 5-HT_{2A} expression on cortical pyramidal cells, note that the 5-HT_{2A} receptor is also expressed on interneurons in the medial temporal lobe (entorhinal cortex, hippocampus, and amygdala) with some estimates being >50% of these neurons (<https://doi.org/10.1016/j.brainresbull.2011.11.006>, <https://doi.org/10.1007/s00221-013-3512-6>, <https://doi.org/10.7554/eLife.66960>, <https://doi.org/10.1016/j.mcn.2008.07.005>, <https://doi.org/10.1038/npp.2008.71>, <https://doi.org/10.1038/s41386-023-01744-8>, <https://doi.org/10.1016/j.brainres.2004.03.016>, [https://doi.org/10.1016/S0022-3565\(24\)37472-5](https://doi.org/10.1016/S0022-3565(24)37472-5), <https://doi.org/10.1002/hipo.22611>, <https://doi.org/10.1016/j.neuron.2024.08.016>). However, with ~1:4 ratio of inhibitory to excitatory neurons in the brain (<https://doi.org/10.1101/2024.09.24.614724>), this can make it seem as if 5-HT_{2A} expression is negligible in the MTL. I think it might be important to mention these receptors, as this manuscript discusses replay.

I see now that Figure 1 mentions that PV cells also express 5-HT_{2A} receptors. This should probably be mentioned earlier.

(2) Page 1, Introduction, "They have further been used for millennia as medicine and in religious rituals..." This might be a romanticization of psychedelics and indigenous groups, as anthropological evidence suggests that intentional psychedelic use might actually be more recent (see work by Manvir Singh and Andy Letcher).

(3) When discussing oneirogens, it could be worth differentiating psychedelics from kappa opioid agonists such as ibogaine and salvinorin A, another class of hallucinogens that some refer to as "oneirogens" (similar to how "psychedelic" is the colloquial term for 5-HT_{2A} agonists). Note that studies have found the effects of *Salvia divinorum* (which contains salvinorin A) to be described more similarly to dreams than psychedelics (<https://doi.org/10.1007/s00213-011-2470-6>). This makes me wonder why the present study is more applicable to 5-HT_{2A} psychedelics than other kappa opioid agonists or other classes of hallucinogens (e.g., NMDA antagonists, muscarinic antagonists, GABA_A agonists).

(4) Page 2, Introduction, "Replay sequences have been shown to be important for learning during sleep [14, 15, 16, 17, 18]: we propose that mechanisms supporting replay-dependent learning during sleep are key to explaining the increases in plasticity caused by psychedelic drug administration." I'm not sure I follow the logic of this point. Dreams happen during REM sleep, whereas replay is most prominent during non-REM sleep. Moreover, while it's not clear what psychedelics do to hippocampal function, most evidence would suggest they impair it. As mentioned, most 5-HT_{2A} receptors in the hippocampus seem to be on inhibitory neurons, and human and animal work finds that psychedelics impair hippocampal-dependent memory encoding (<https://doi.org/10.1037/rev0000455>, <https://doi.org/10.1037/rev0000455>, <https://doi.org/10.3389/fnbeh.2014.00180>, <https://doi.org/10.1002/hipo.22712>). One study even found that psilocin impairs hippocampal-dependent memory retrieval (<https://doi.org/10.3389/fnbeh.2014.00180>). Note that this is all in reference to the acute effects (psychedelics may post-acutely enhance hippocampal-dependent memory, <https://doi.org/10.1007/s40265-024-02106-4>).

(5) Page 2, Introduction, "In total, our model of the functional effect of psychedelics on pyramidal neurons could provide a explanation for the perceptual psychedelic experience in terms of learning mechanisms for consolidation during sleep..." In contrast to my previous point, I think this could be possible. Three datasets have found that psychedelics may enhance cortical-dependent memory encoding (i.e., familiarity; <https://doi.org/10.1037/rev0000455>, <https://doi.org/10.1037/rev0000455>), and two studies found that post-encoding administration of psychedelics retroactively enhanced memory that may be less hippocampal-dependent/more cortical-dependent (<https://doi.org/10.1016/j.neuropharm.2012.06.007>, <https://doi.org/10.1016/j.euroneuro.2022.01.114>). Moreover, and as mentioned below, 5 studies have found decoupling between the hippocampus and the cortex (<https://doi.org/10.3389/fnhum.2014.00020>, <https://doi.org/10.1002/hbm.22833>, <https://doi.org/10.1016/j.celrep.2021.109714>, https://doi.org/10.1162/netn_a_00349, <https://doi.org/10.1038/s41586-024-07624-5>), something potentially also observed during REM sleep that is thought to support consolidation (<https://doi.org/10.1073/pnas.2123432119>). These findings should probably be discussed.

(6) Page 2, Introduction, "In this work, we show that within a neural network trained via Wake-Sleep, it is possible to model the action of classical psychedelics (i.e. 5-HT_{2a} receptor agonism)..." Note that 5-HT_{2A} agonism alone is not sufficient to explain the effects of psychedelics, given that there are 5-HT_{2A} agonists that are non-hallucinogenic (e.g., lisuride).

(7) Page 2, Introduction, "...by shifting the balance during the wake state from the bottom-up pathways to the top-down pathways, thereby making the 'wake' network states more 'dream-like'." I could have included this in the previous point, but I felt that this idea deserved its own point. There has been a rather dogmatic assertion that psychedelics diminish top-down processing and/or enhance bottom-up processing, and I appreciate that the authors have not accepted this as fact. However, because this is an unfortunately prominent idea, I think it ought to be fleshed out more by first mentioning that it's one of the tenets of REBUS. REBUS has become a popular model of psychedelic drug action, but it's largely unfalsifiable (it's based on two unfalsifiable models, predictive processing and integrated information theory), so the findings from this study could tighten it up a bit. Second, there have now been a handful of studies that have attempted to study directionality in information flow under psychedelics, and the findings are rather mixed including increased bottom-up/decreased top-down effects (<https://doi.org/10.7554/eLife.59784>, <https://doi.org/10.1073/pnas.1815129116>; note that the latter "bottom-up" effect involves subcortical-cortical connections in which it's less clear what's actually "higher-/lower-level"), increased top-down/decreased bottom-up effects (<https://doi.org/10.1038/s41380-024-02632-3>, <https://doi.org/10.1016/j.euroneuro.2016.03.018>), or both (<https://doi.org/10.1016/j.neuroimage.2019.116462>, <https://doi.org/10.1016/j.neuropharm.2017.10.039>), though most of these studies are aggregating across largely

inhomogeneous states (i.e., resting-state). Lastly, and somewhat problematically, facilitated top-down processing is also an idea proposed in psychosis that's based partially on findings with acute ketamine administration (note that all hallucinations to some degree might rely on top-down facilitation, as a hallucination involves a high-level concept that impinges on lower-level sensory areas; see work by Phil Corlett). While psychosis and the effects of ketamine have some similarities with psychedelics, there are certainly differences, and I think the goal of this manuscript is to uniquely describe 5-HT_{2A} psychedelics (again, I'm left wondering why tweaking α in the Wake-Sleep algorithm is any more applicable to psychedelics than other hallucinogenic conditions).

(8) Figure 2 equates α with a "psychedelic dose," but this is a bit misleading, as neither the algorithm nor an individual was administered a psychedelic. α is instead a hypothetical proxy for a psychedelic dose. Moreover, if the model were recapitulating the effects of psychedelics, shouldn't these images look more psychedelic as α increases (e.g., they may look like images put through the DeepDream algorithm).

(9) Page 11, Methods, "...and the gate α ensures that learning only occurs during sleep mode... The $(1 - \alpha)$ gate in this case ensures that plasticity only occurs during the Wake mode." Much of the math escapes me, so perhaps I'm misunderstanding these statements, but learning and plasticity certainly happen during both wake and sleep, making me wonder what is meant by these statements. Moreover, if plasticity is simply neural changes, couldn't plasticity be synonymous with neural learning? Perhaps plasticity and learning are meant to refer to different types of neural changes. It might be worth clarifying this, as a general problem in psychedelic research is that psychedelics are described as facilitating plasticity when brains are changing at every moment (hence not experiencing every moment as the same), and psychedelics don't impact all forms of plasticity equally. For example, psychedelics may not necessarily enhance neurogenesis or the addition of certain receptor types, and they impair certain forms of learning (i.e., episodic memory encoding). What is typically meant by plasticity enhancements induced by psychedelics (and where there's the most evidence) is dendritic plasticity (i.e., the growth of dendrites and spines). Whatever is meant by "plasticity" should be clarified in its first instance in this manuscript.

(10) Page 12, Methods, "During training, neural network activity is either dominated entirely by bottom-up inputs (Wake, $\alpha = 0$) or by top-down inputs (Sleep, $\alpha = 1$)." Again, I could be misunderstanding the mathematical formulation, but top-down inputs operate during wake, and bottom-up inputs can operate during sleep (people can wake up or even incorporate noise from their environments into sleep).

(11) Page 4, Results, "Thus, we can capture the core idea behind the oneirogen hypothesis using the Wake-Sleep algorithm, by postulating that the bottom-up basal synapses are predominantly driving neural activity during the Wake phase (when α is low)." However, several pieces of evidence (and the first circuit model of psychedelic drug action) suggest that psychedelics enhance functional connectivity and potentially even effective connectivity from the thalamus to the cortex (<https://doi.org/10.1093/brain/awab406>). Note that psychedelics may not equally impact all subcortical structures. REBUS proposes the opposite of the current study, that psychedelics facilitate bottom-up information flow, with one of the few explicit predictions being that psychedelics should facilitate information flow from the hippocampus to the default mode network. However, as mentioned earlier, 5 studies have found that psychedelics diminish functional connectivity between the hippocampus and cortex (including the DMN but also V1).

(12) Page 4, Results, "...and have an excitatory effect that positively modulates glutamatergic transmission..." Note that this may not be brainwide. While psychedelics were found to increase glutamatergic transmission in the cortex, they were also found to decrease hippocampal glutamate (consistent with inhibition of the hippocampus, <https://doi.org/10.1038/s41386-020-0718-8>).

(13) Page 5, "...which are similar to the 'breathing' and 'rippling' phenomena reported by psychedelic drug users at low doses..." Although it's sometimes unclear what is meant by "low doses," the breathing/rippling effect of psychedelics occurs at moderate and high doses as well.

(14) I watched the videos, and it's hard for me to say there was some stark resemblance to psychedelic imagery. In contrast, for example, when the DeepDream algorithm came out, it did seem to capture something quite psychedelic.

(15) Page 5, "This form of strongly correlated tuning has been observed in both cortex and the hippocampus." If this has been observed under non-psychedelic conditions, what does this tell us about this supposed model of psychedelics?

(16) Page 6, with regards to neural variability, "...but whether this phenomenon [increased variability] is general across tasks and cortical areas remains to be seen." First, is variability here measured as variance? In fMRI datasets that have been used to support the Entropic Brain Hypothesis, note that variance tends to decrease, though certain measures of entropy increase (e.g., Figure 4A here <https://doi.org/10.1073/pnas.1518377113> shows global variance decreases, and this reanalysis of those data <https://doi.org/10.1002/hbm.23234> finds some entropy increases). Thus, variance and entropy should not be confused (in theory, one could cycle through several more brain states that are however, similar to each other, which would produce more entropy with decreased variance). Second, and perhaps more problematically for the EBH, is that the entropy effects of psychedelics completely disappear when one does a task, and unfortunately, the authors of these findings have misinterpreted them. What they'll say is that engaging in boring cognitive tasks or watching a video decreases entropy under psychedelics, but what you can see in Figure 1b of <https://doi.org/10.1021/acscchemneuro.3c00289> and Figure 4b of <https://doi.org/10.1038/s41586-024-07624-5> is that entropy actually increases under sober conditions when you do a task. That is, it's a rather boring finding. Essentially, when resting in a scanner while sober, many may actually rest (including falling asleep, especially when subjects are asked to keep their eyes closed), and if you perform a task, brain activity should become more complex relative to doing nothing/falling asleep. When under a psychedelic, one can't fall asleep and thus, there's less change (though note that both of the above studies found numerical increases when performing tasks). Lastly, again I should note that the findings of the present study actually go against EBH/REBUS, given that the findings are increased top-down effects when EBH/REBUS predicts decreased top-down/increased bottom-up effects.

(17) Page 6, "Because psychedelic drug administration increases influence of apical dendritic inputs on neural activity in our model, we found that silencing apical dendritic activity reduced across stimulus neural variability more as the psychedelic drug dose increases." I again want to point out that alpha is not the equivalent of a psychedelic dose here, but rather a parameter in the model that is being proposed as a proxy.

(18) Page 8, "Experimentally, plasticity dynamics which could, theoretically, minimize such a prediction error have been observed in cortex [66, 67], and it has also been proposed that behavioral timescale plasticity in the hippocampus could subserve a similar function [68]. We found that plasticity rules of this kind induce strong correlations between inputs to the apical and basal dendritic compartments of pyramidal neurons, which have been observed in the hippocampus and cortex [55, 56]." Note that the plasticity effects of psychedelics are sometimes not observed in the hippocampus or are even observed as decreases (reviewed in <https://doi.org/10.1038/s41386-022-01389-z>).

(19) Page 9, as is mentioned, REBUS proposes that there should be a decrease in top-down effects under psychedelics, which goes against what is found here, but as I describe above, the effects of psychedelics on various measures of directionality have been quite mixed.

(20) Unless I'm misunderstanding something, it seems to be a bit of a jump to infer that simply changing alpha in your model is akin to psychedelic dosing. Perhaps if the model implemented biologically plausible 5-HT_{2A} expression and/or its behavior were constrained by common features of a psychedelic experience (e.g., fractal-like visuals imposed onto perception, inability to fall asleep, etc.), I'd be more inclined to see the parallels between alpha and psychedelics dosing. However, it would still need to recapitulate unique effects of psychedelics (e.g., impairments in hippocampal-dependent memory with sparing/facilitation of cortical memory). At the moment, it seems like whatever the model is doing is applicable to any hallucinogenic drug or even psychosis.

<https://doi.org/10.7554/eLife.105968.1.sa2>

Reviewer #2 (Public review):

This work is a nice contribution to the literature in articulating a specific, testable theory of how psychedelics act to generate hallucinations and plasticity. The connection to replay, however - including in the title, abstract, and framing throughout the paper - is not well fleshed out.

In particular, the paper's framing seems to conflate replay, dreams, and top-down processing, but these are not one and the same. Picard-Delano et al. TICS 2023 provides a useful review of the differences between replay and dreams. One key point is that most replay has been observed during NREM sleep, but our canonically bizarre / vivid dreams occur during REM. Top-down connections have also been proposed to be used for many processes aside from replay. The paper would benefit from much more precision and nuance on these points.

I believe the paper is missing demonstrations or speculation about how plasticity under various doses of psychedelics relates to changes in performance, which would be an important link to the replay-dependent learning literature.

Are there renderings available for 'ripple' effects of psychedelics that could be included, to allow readers to compare the model's hallucinations to humans? Short of this, it would be useful to have a more detailed description of what rippling is. (For those readers without firsthand knowledge!) It is currently difficult to assess how close the match is.

<https://doi.org/10.7554/eLife.105968.1.sa1>

Author response:

We thank the reviewers for the valuable and constructive reviews. Thanks to these, we believe the article will be considerably improved. We have organized our response to address points that are relevant to both reviewers first, after which we address the unique concerns of each individual reviewer separately. We briefly paraphrase each concern and provide comments for clarification, outlining the precise changes that we will make to the text.

Common Concerns (Reviewer 1 & Reviewer 2):

Can you clarify how NREM and REM sleep relate to the oneirogen hypothesis?

Within the submission draft we tried to stay agnostic as to whether mechanistically similar replay events occur during NREM or REM sleep; however, upon a more thorough literature review, we think that there is moderately greater evidence in favor of Wake-Sleep-type replay occurring during REM sleep which is related to classical psychedelic drug mechanisms of action.

First, we should clarify that replay has been observed during both REM and NREM sleep, and dreams have been documented during both sleep stages, though the characteristics of dreams differ across stages, with NREM dreams being more closely tied to recent episodic experience and REM dreams being more bizarre/hallucinatory (see Stickgold et al., 2001 for a review). Replay during sleep has been studied most thoroughly during NREM sharp-wave ripple events, in which significant cortical-hippocampal coupling has been observed (Ji & Wilson, 2007). However, it is critical to note that the quantification methods used to identify replay events in the hippocampal literature usually focus on identifying what we term ‘episodic replay,’ which involves a near-identical recapitulation of neural trajectories that were recently experienced during waking experimental recordings (Tingley & Peyrarch, 2020). In contrast, our model focuses on ‘generative replay,’ where one expects only a statistically similar reproduction of neural activity, without any particular bias towards recent or experimentally controlled experience. This latter form of replay may look closer to the ‘reactivation’ observed in cortex by many studies (e.g. Nguyen et al., 2024), where correlation structures of neural activity similar to those observed during stimulus-driven experience are recapitulated. Under experimental conditions in which an animal is experiencing highly stereotyped activity repeatedly, over extended periods of time, these two forms of replay may be difficult to dissociate.

Interestingly, though NREM replay has been shown to couple hippocampal and cortical activity, a similar study in waking animals administered psychedelics found hippocampal replay without any obvious coupling to cortical activity (Domenico et al., 2021). This could be because the coupling was not strong enough to produce full trajectories in the cortex (psychedelic administration did not increase ‘alpha’ enough), and that a causal manipulation of apical/basal influence in the cortex may be necessary to observe the increased coupling. Alternatively, as Reviewer 1 noted, it may be that psychedelics induce a form of hippocampus-decoupled replay, as one would expect from the REM stage of a recently proposed complementary learning systems model (Singh et al., 2022).

Evidence in favor of a similarity between the mechanism of action of classical psychedelics and the mechanism of action of memory consolidation/learning during REM sleep is actually quite strong. In particular, studies have shown that REM sleep increases the activity of soma-targeting parvalbumin (PV) interneurons and decreases the activity of apical dendrite-targeting somatostatin (SOM) interneurons (Niethard et al., 2021), that this shift in balance is controlled by higher-order thalamic nuclei, and that this shift in balance is critical for synaptic consolidation of both monocular deprivation effects in early visual cortex (Zhou et al., 2020) and for the consolidation of auditory fear conditioning in the dorsal prefrontal cortex (Aime et al., 2022). These last studies were not discussed in the present manuscript—we will add them, in addition to a more nuanced description of the evidence connecting our model to NREM and REM replay.

Can you explain how synaptic plasticity induced by psychedelics within your model relates to learning at a behavioral level?

While the Wake-Sleep algorithm is a useful model for unsupervised statistical learning, it is not a model of reward or fear-based conditioning, which likely occur via different mechanisms in the brain (e.g. dopamine-dependent reinforcement learning or serotonin-dependent emotional learning). The Wake-Sleep algorithm is a ‘normative plasticity algorithm,’ that connects synaptic plasticity to the formation of structured neural representations, but it is not the case that all synaptic plasticity induced by psychedelic administration within our model should induce beneficial learning effects. According to the Wake-Sleep algorithm, plasticity at apical synapses is enhanced during the Wake phase, and plasticity at basal synapses is enhanced during the Sleep phase; under the oneirogen hypothesis, hallucinatory conditions (increased ‘alpha’) cause an increase in plasticity at both

apical and basal sites. Because neural activity is in a fundamentally aberrant state when ‘alpha’ is increased, there are no theoretical guarantees that plasticity will improve performance on any objective: psychedelic-induced plasticity within our model could perhaps better be thought of as ‘noise’ that may have a positive or negative effect depending on the context.

In particular, such ‘noise’ may be beneficial for individuals or networks whose synapses have become locked in a suboptimal local minimum. The addition of large amounts of random plasticity could allow a system to extricate itself from such local minima over subsequent learning (or with careful selection of stimuli during psychedelic experience), similar to simulated annealing optimization approaches. If our model were fully validated, this view of psychedelic-induced plasticity as ‘noise’ could have relevance for efforts to alleviate the adverse effects of PTSD, early life trauma, or sensory deprivation; it may also provide a cautionary note against repeated use of psychedelic drugs within a short time frame, as the plasticity changes induced by psychedelic administration under our model are not guaranteed to be good or useful in-and-of themselves without subsequent re-learning and compensation.

We should also note that we have deliberately avoided connecting the oneirogen hypothesis model to fear extinction experimental results that have been observed through recordings of the hippocampus or the amygdala (Bombardi & Giovanni, 2013; Jiang et al., 2009; Kelly et al., 2024; Tiwari et al., 2024). Both regions receive extensive innervation directly from serotonergic synapses originating in the dorsal raphe nucleus, which have been shown to play an important role in emotional learning (Lesch & Waider, 2012); because classical psychedelics may play a more direct role in modulating this serotonergic innervation, it is possible that fear conditioning results (in addition to the anxiolytic effects of psychedelics) cannot be attributed to a shift in balance between apical and basal synapses induced by psychedelic administration. We will provide a more detailed review of these results in the text, as well as more clarity regarding their relation to our model.

Reviewer 1 Concerns:

Is it reasonable to assign a scalar parameter ‘alpha’ to the effects of classical psychedelics? And is your proposed mechanism of action unique to classical psychedelics? E.g. Could this idea also apply to kappa opioid agonists, ketamine, or the neural mechanisms of hallucination disorders?

We will clarify that within our model ‘alpha’ is a parameter that reflects the balance between apical and basal synapses in determining the activity of neurons in the network. For the sake of simplicity we used a single ‘alpha’ parameter, but realistically, each neuron would have its own ‘alpha’ parameter, and different layers or individual neurons could be affected differentially by the administration of any particular drug; therefore, our scalar ‘alpha’ value can be thought of as a mean parameter for all neurons, disregarding heterogeneity across individual neurons.

There are many different mechanisms that could theoretically affect this ‘alpha’ parameter, including: 5-HT_{2a} receptor agonism, kappa opioid receptor binding, ketamine administration, or possibly the effects of genetic mutations underlying the pathophysiology of complex developmental hallucination disorders. We focused exclusively on 5-HT_{2a} receptor agonism for this study because the mechanism is comparatively simple and extensively characterized, but similar mechanisms may well be responsible for the hallucinatory symptoms of a variety of drugs and disorders.

Can you clarify the role of 5-HT_{2a} receptor expression on interneurons within your model?

While we mostly focused on the effects of 5-HT_{2a} receptors on the apical dendrites of pyramidal neurons, these receptors are also expressed on soma-targeting parvalbumin (PV) interneurons. This expression on PV interneurons is consistent with our proposed psychedelic mechanism of action, because it could lead to a coordinated decrease in the influence of somatic and proximal dendritic inputs while increasing the influence of apical dendritic inputs. We will elaborate on this point, and will move the discussion earlier in the text.

Discussions of indigenous use of psychedelics over millenia may amount to over-romanticization.

We will take great care to conduct a more thorough literature review to reevaluate our statement regarding indigenous psychedelic use (including the citations you suggested), and will either provide a more careful statement or remove this discussion from our introduction entirely, as it has little bearing on the rest of the text. The Ethics Statement will also be modified accordingly.

You isolate the 5-HT_{2a} agonism as the mechanism of action underlying 'alpha' in your model, but there exist 5-HT_{2a} agonists that do not have hallucinatory effects (e.g. lisuride). How do you explain this?

Lisuride has much-reduced hallucinatory effects compared to other psychedelic drugs at clinical doses (though it does indeed induce hallucinations at high doses; Marona-Lewicka et al., 2002), and we should note that serotonin (5-HT) itself is pervasive in the cortex without inducing hallucinatory effects during natural function. Similarly, MDMA is a partial agonist for 5-HT_{2a} receptors, but it has much-reduced perceptual hallucination effects relative to classical psychedelics (Green et al., 2003) in addition to many other effects not induced by classical psychedelics.

Therefore, while we argue that 5-HT_{2a} agonism induces an increase in influence of apical dendritic compartments and a decrease in influence of basal/somatic compartments, and that this change induces hallucinations, we also note that there are many other factors that control whether or not hallucinations are ultimately produced, so that not all 5-HT_{2a} agonists are hallucinogenic. We will discuss two such factors in our revision: 5-HT receptor binding affinity and cellular membrane permeability.

Importantly, many 5-HT_{2a} receptor agonists are also 5-HT_{1a} receptor agonists (e.g. serotonin itself and lisuride), while MDMA has also been shown to increase serotonin, norepinephrine, and dopamine release (Green et al., 2003). While 5-HT_{2a} receptor agonism has been shown to reduce sensory stimulus responses (Michaël et al., 2019), 5-HT_{1a} receptor agonism inhibits spontaneous cortical activity (Azimi et al., 2020); thus one might expect the net effect of administering serotonin or a nonselective 5-HT receptor agonist to be widespread inhibition of a circuit, as has been observed in visual cortex (Azimi et al., 2020). Therefore, selective 5-HT_{2a} agonism is critical for the induction of hallucinations according to our model, though any intervention that jointly excites pyramidal neurons' apical dendrites and inhibits their basal/somatic compartments across a broad enough area of cortex would be predicted to have a similar effect. Lisuride has a much higher binding affinity for 5-HT_{1a} receptors than, for instance, LSD (Marona-Lewicka et al., 2002).

Secondly, it has recently been shown that both the head-twitch effect (a coarse behavioral readout of hallucinations in animals) and the plasticity effects of psychedelics are abolished when administering 5-HT_{2a} agonists that are impermeable to the cellular membrane because of high polarity, and that these effects can be rescued by temporarily rendering the cellular membrane permeable (Vargas et al., 2023). This suggests that the critical hallucinatory effects

of psychedelics (apical excitation according to our model) may be mediated by intracellular 5-HT_{2A} receptors. Notably, serotonin itself is not membrane permeable in the cortex.

Therefore, either of these two properties could play a role in whether a given 5-HT_{2A} agonist induces hallucinatory effects. We will provide a considerably extended discussion of these nuances in our revision.

Your model proposes that an increase in top-down influence on neural activity underlies the hallucinatory effects of psychedelics. How do you explain experimental results that show increases in bottom-up functional connectivity (either from early sensory areas or the thalamus)?

Firstly, we should note that our proposed increase in top-down influence is a causal, biophysical property, not necessarily a statistical/correlative one. As such, we will stress that the best way to test our model is via direct intervention in cortical microcircuitry, as opposed to correlative approaches taken by most fMRI studies, which have shown mixed results with regard to this particular question. Correlative approaches can be misleading due to dense recurrent coupling in the system, and due to the coarse temporal and spatial resolution provided by noninvasive recording technologies (changes in statistical/functional connectivity do not necessarily correspond to changes in causal/mechanistic connectivity, i.e. correlation does not imply causation).

There are two experimental results that appear to contradict our hypothesis that deserve special consideration in our revision. The first shows an increase in directional thalamic influence on the distributed cortical networks after psychedelic administration (Preller et al., 2018). To explain this, we note that this study does not distinguish between lower-order sensory thalamic nuclei (e.g. the lateral and medial geniculate nuclei receiving visual and auditory stimuli respectively) and the higher-order thalamic nuclei that participate in thalamocortical connectivity loops (Whyte et al., 2024). Subsequent more fine-grained studies have noted an increase in influence of higher order thalamic nuclei on the cortex (Pizzi et al., 2023; Gaddis et al., 2022), and in fact extensive causal intervention research has shown that classical psychedelics (and 5-HT_{2A} agonism) decrease the influence of incoming sensory stimuli on the activity of early sensory cortical areas, indicating decoupling from the sensory thalamus (Evarts et al., 1955; Azimi et al., 2020; Michaeli et al. 2019). The increased influence of higher-order thalamic nuclei is consistent with both the cortico-striatal-thalamo-cortical (CTSC) model of psychedelic action as well as the oneirogen hypothesis, since higher-order thalamic inputs modulate the apical dendrites of pyramidal neurons in cortex (Whyte et al., 2024).

The second experimental result notes that DMT induces traveling waves during resting state activity that propagate from early visual cortex to deeper cortical layers (Alamia et al., 2020). There are several possibilities that could explain this phenomenon: 1) it could be due to the aforementioned difficulties associated with directed functional connectivity analyses, 2) it could be due to a possible high binding affinity for DMT in the visual cortex relative to other brain areas, or 3) it could be due to increases in apical influence on activity caused by local recurrent connectivity within the visual cortex which, in the absence of sensory input, could lead to propagation of neural activity from the visual cortex to the rest of the brain. This last possibility is closest to the model proposed by (Ermentrout & Cowan, 1979), and which we believe would be best explained within our framework by a topographically connected recurrent network architecture trained on video data; a potentially fruitful direction for future research.

Shouldn't the hallucinations generated by your model look more 'psychedelic,' like those produced by the DeepDream algorithm?

We believe that the differences in hallucination visualization quality between our algorithm and DeepDream are mostly due to differences in the scale and power of the models used across these two studies. We are confident that with more resources (and potentially theoretical innovations to improve the Wake-Sleep algorithm's performance) the produced hallucination visualizations could become more realistic, but we believe this falls outside the scope of the present study.

We note that more powerful generative models trained with backpropagation are able to produce surreal images of comparable quality (Rezende et al., 2014; Goodfellow et al., 2020; Vahdat & Kautz, 2020), though these have not yet been used as a model of psychedelic hallucinations. However, the DeepDream model operates on top of large pretrained image processing models, and does not provide a biologically mechanistic/testable interpretation of its hallucination effects. When training smaller models with a local synaptic plasticity rule (as opposed to backpropagation), the hallucination effects are less visually striking due to the reduced quality of our trained generative model, though they are still strongly tied to the statistics of sensory inputs, as quantified by our correlation similarity metric (Fig. 5b). We will provide a more detailed explanation of this phenomenon when we discuss our model limitations in our revised manuscript.

Your model assumes domination by entirely bottom-up activity during the 'wake' phase, and domination entirely by top-down activity during 'sleep,' despite experimental evidence indicating that a mixture of top-down and bottom-up inputs influence neural activity during both stages in the brain. How do you explain this?

Our use of the Wake-Sleep algorithm, in which top-down inputs (Sleep) or bottom-up inputs (Wake) dominate network activity is an over-simplification made within our model for computational and theoretical reasons. Models that receive a mixture of top-down and bottom-up inputs during 'Wake' activity do exist (in particular the closely related Boltzmann machine (Ackley et al., 1985)), but these models are considerably more computationally costly to train due to a need to run extensive recurrent network relaxation dynamics for each input stimulus. Further, these models do not generalize as cleanly to processing temporal inputs. For this reason, we focused on the Wake-Sleep algorithm, at the cost of some biological realism, though we note that our model should certainly be extended to support mixed apical-basal waking regimes. We will make sure to discuss this in our 'Model Limitations' section.

Your model proposes that 5-HT_{2A} agonism enhances glutamatergic transmission, but this is not true in the hippocampus, which shows decreases in glutamate after psychedelic administration.

We should note that our model suggests only compartment specific increases in glutamatergic transmission; as such, our model does not predict any particular directionality for measures of glutamatergic transmission that includes signaling at both apical and basal compartments in aggregate, as was measured in the provided study (Mason et al., 2020).

You claim that your model is consistent with the Entropic Brain theory, but you report increases in variance, not entropy. In fact, it has been shown that variance decreases while entropy increases under psychedelic administration. How do you explain this discrepancy?

Unfortunately, 'entropy' and 'variance' are heavily overloaded terms in the noninvasive imaging literature, and the particularities of the method employed can exert a strong influence on the reported effects. The reduction in variance reported by (Carhart-Harris et al., 2016) is a very particular measure: they are reporting the variance of resting state

synchronous activity, averaged across a functional subnetwork that spans many voxels; as such, the reduction in variance in this case is a reduction in broad, synchronous activity. We do not have any resting state synchronous activity in our network due to the simplified nature of our model (particularly an absence of recurrent temporal dynamics), so we see no reduction in variance in our model due to these effects.

Other studies estimate ‘entropy’ or network state disorder via three different methods that we have been able to identify. 1) (Carhart-Harris et al., 2014) uses a different measure of variance: in this case, they subtract out synchronous activity within functional subnetworks, and calculate variability across units in the network. This measure reports increases in variance (Fig. 6), and is the closest measure to the one we employ in this study. 2) (Lebedev et al., 2016) uses sample entropy, which is a measure of temporal sequence predictability. It is specifically designed to disregard highly predictable signals, and so one might imagine that it is a measure that is robust to shared synchronous activity (e.g. resting state oscillations). 3) (Mediano et al., 2024) uses Lempel-Ziv complexity, which is, similar to sample entropy, a measure of sequence diversity; in this case the signal is binarized before calculation, which makes this method considerably different from ours. All three of the preceding methods report increases in sequence diversity, in agreement with our quantification method. Our strongest explanation for why the variance calculation in (Carhart-Harris et al., 2016) produces a variance reduction is therefore due to a reduction in low-rank synchronous activity in subnetworks during resting state.

As for whether the entropy increase is meaningful: we share Reviewer 1’s concern that increases in entropy could simply be due to a higher degree of cognitive engagement during resting state recordings, due to the presence of sensory hallucinations or due to an inability to fall asleep. This could explain why entropy increases are much more minimal relative to non-hallucinating conditions during audiovisual task performance (Siegel et al., 2024; Mediano et al., 2024). However, we can say that our model is consistent with the Entropic Brain Theory without including any form of ‘cognitive processing’: we observe increases in variability during resting state in our model, but we observe highly similar distributions of activity when averaging over a wide variety of sensory stimulus presentations (Fig. 5b-c). This is because variability in our model is not due to unstructured noise: it corresponds to an exploration of network states that would ordinarily be visited by some stimulus. Therefore, when averaging across a wide variety of stimuli, the distribution of network states under hallucinating or non-hallucinating conditions should be highly similar.

One final point of clarification: here we are distinguishing Entropic Brain Theory from the REBUS model—the oneirogen hypothesis is consistent with the increase in entropy observed experimentally, but in our model this entropy increase is not due to increased influence of bottom-up inputs (it is due instead to an increase in top-down influence). Therefore, one could view the oneirogen hypothesis as consistent with EBT, but inconsistent with REBUS.

You relate your plasticity rule to behavioral-timescale plasticity (BTSP) in the hippocampus, but plasticity has been shown to be reduced in the hippocampus after psychedelic administration. Could you elaborate on this connection?

When we were establishing a connection between our ‘Wake-Sleep’ plasticity rule and BTSP learning, the intended connection was exclusively to the mathematical form of the plasticity rule, in which activity in the apical dendrites of pyramidal neurons functions as an instructive signal for plasticity in basal synapses (and vice versa): we will clarify this in the text. Similarly, we point out that such a plasticity rule tends to result in correlated tuning between apical and basal dendritic compartments, which has been observed in hippocampus and cortex: this is intended as a sanity check of our mapping of the Wake-Sleep algorithm to cortical microcircuitry, and has limited further bearing on the effects of psychedelics specifically.

Reduction in plasticity in the hippocampus after psychedelic administration could be due to a complementary learning systems-type model, in which the hippocampus becomes partly decoupled from the cortex during REM sleep (Singh et al., 2022); were this to be the case, it would not be incompatible with our model, which is mostly focused on the cortex. Notably, potentiating 5HT-2a receptors in the ventral hippocampus does not induce the head-twitch response, though it does produce anxiolytic effects (Tiwari et al., 2024), indicating that the hallucinatory and anxiolytic effects of classical psychedelics may be partly decoupled.

Reviewer 2 Concerns:

Could you provide visualizations of the ‘ripple’ phenomenon that you’re referring to?

We will do this! For now, you can get a decent understanding of what the ‘ripple effect’ looks like from the ‘eyes closed’ hallucination condition for networks trained on CIFAR10 (Fig. 2d). The ripple effect that we are referring to is very similar, except it is superimposed on a naturalistic image under ordinary viewing conditions; to give a higher quality visualization of the ripple phenomenon itself, we will subtract out the static contribution of the image itself, leaving only the ripple phenomenon.

Could you provide a more nuanced description of alternative roles for top-down feedback, beyond being used exclusively for learning as depicted in your model?

For the sake of simplicity, we only treat top-down inputs in our model as a source of an instructive teaching signal, the originator of generative replay events during the Sleep phase, and as the mechanism of hallucination generation. However, as discussed in a response to a previous question, in the cortex pyramidal neurons receive and respond to a mixture of top-down and bottom-up processing.

There are a variety of theories for what role top-down inputs could play in determining network activity. To name several, top-down input could function as: 1) a denoising/pattern completion signal (Kadkhodaie & Simoncelli, 2021), 2) a feedback control signal (Podlaski & Machens, 2020), 3) an attention signal (Lindsay, 2020), 4) ordinary inputs for dynamic recurrent processing that play no specialized role distinct from bottom-up or lateral inputs except to provide inputs from higher-order association areas or other sensory modalities (Kar et al., 2019; Tugsbayar et al., 2025). Though our model does not include these features, they are perfectly consistent with our approach.

In particular, denoising/pattern completion signals in the predictive coding framework (closely related to the Wake-Sleep algorithm) also play a role as an instructive learning signal (Salvatori et al., 2021); and top-down control signals can play a similar role in some models (Gilra & Gerstner, 2017; Meulemans et al., 2021). Thus, options 1 and 2 are heavily overlapping with our approach, and are a natural consequence of many biologically plausible learning algorithms that minimize a variational free energy loss (Rao & Ballard, 1997; Ackley et al., 1985). Similarly, top-down attentional signals can exist alongside top-down learning signals, and some models have argued that such signals can be heavily overlapping or mutually interchangeable (Roelfsema & van Ooyen, 2005). Lastly, generic recurrent connectivity (from any source) can be incorporated into the Wake-Sleep algorithm (Dayan & Hinton, 1996), though we avoided doing this in the present study due to an absence of empirical architecture exploration in the literature and the computational complexity associated with training on time series data.

To conclude, there are a variety of alternative functions proposed for top-down inputs onto pyramidal neurons in the cortex, and we view these additional features as mutually compatible with our approach; for simplicity we did not include them in our model, but we

believe that these features are unlikely to interfere with our testable predictions or empirical results.

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